NMR 谱图的采集与处理

Acquisition and Processing of NMR Spectra

CHMD11 讲义 86 页 → 中英对照速学版：一页一个概念，左边看图，右边中英逐句对齐

5 分钟	只看前 3 页：全局地图、取舍总表、必背公式与数字
1 小时	读第 1–6 章。每页先读「一句话」，再看图，最后看要点
考前 15 分钟	第 7 章速查表 + 第 8 章自测题

① 上机准备 Prepare　让样品和硬件就绪　get sample and hardware ready　第 5 章 · p.27

样品 Sample → 调谐匹配 Tune & match → 锁场 Lock → 匀场 Shim → 校准 p1 Calibrate → 设参数 Set parameters

② 采集 Acquire　把微伏级振荡电流变成一串数字　turn a μV oscillation into numbers　第 1–4 章 · p.4

脉冲 RF pulse → 线圈信号 Coil signal, μV → 前置放大 Pre-amp, RG → ADC 数字化 Digitize → FID TD points → 重复 NS 次 × NS scans

③ 处理 Process　把时域数字变成可读的谱图　turn time-domain data into a spectrum　第 6 章 · p.33

线性预测 LP, optional → 窗函数 Window, LB → 补零 ZF, SI → 傅里叶变换 FT → 相位 PH0, PH1 → 基线 Baseline → 定标 Reference, SR

整份讲义就三句话 The lecture in three sentences

信号极小而且是模拟的。先放大、再数字化，这两步都不能把信息弄丢。

The signal is tiny and analogue. Amplify it, then digitize it, without losing information at either step.

每个参数都是取舍。灵敏度、分辨率、定量准确度、实验时间，四者互相交换。

Every parameter is a trade-off among sensitivity, resolution, quantitative accuracy and time.

处理能把数据修好看，但造不出真实信息。补零不加信息，线性预测加的是“假”数据。

Processing can polish data but cannot create real information. Zero filling adds none; linear prediction adds “fake” data.

标签：**一句话** 本页核心 · **出错** 设错时的样子 · **记忆钩子** 比喻和口诀 · **讲义外补充** 讲义没写、我补的解释 · **讲义勘误** 讲义的小错 · 图注右下角标明来自讲义哪页，或是数值模拟。
章节顺序跟讲义一致：先懂参数，第 5 章上机时才看得懂 p1、O1P。

取舍总表 The trade-off table

记住这张表，就记住了讲义的骨架：每个旋钮往哪拧，赚什么，赔什么。
The skeleton of the lecture: for each knob, what you gain and what you pay.

参数 Parameter	调高赚什么 Gain	代价 Cost	太低 / 设错 Too low	太高 Too high	经验值 Rule of thumb
TD	分辨率 ↑ Resolution up	AQ 更长，更费时 Longer AQ, more time	FID 被截断：波纹、分辨率差、定量差 Truncation: wiggles, poor resolution, lost quantitation	多花时间 Wasted time	FID 在 AQ 内衰减完 FID decays within AQ
SW	盖住更宽的范围 Covers more	TD 不变时 AQ ↓、分辨率 ↓ At fixed TD: shorter AQ, lower resolution	峰落在窗外；F1 维混叠 Peaks missed; aliasing in F1	点数浪费在空白区 Points spent on empty regions	500 MHz：^{1}H 8–10 kHz，^{13}C 约 25 kHz At 500 MHz: ^{1}H 8–10 kHz, ^{13}C ~25 kHz
RG	小峰信号更好 Better signal from small peaks	有溢出风险 Risk of overflow	不失真，但白丢 SNR No distortion, but SNR thrown away	ADC 溢出：谱图畸变、假峰 ADC overflow: distortion, fake peaks	单扫下调到无畸变，或 rga；碳谱用最大值 Lower until clean, or rga; max for carbon
D1	定量更准 Better quantitation	实验更慢 Longer experiment	磁化恢复不全，强度下降 Incomplete recovery, lower intensity	多花时间 Wasted time	1D：3–5 × T1；2D：1–3 × T1 1D: 3–5 × T1; 2D: 1–3 × T1
NS	SNR ∝ √NS SNR scales with √NS	时间 ∝ NS Time scales with NS	噪声大，小峰看不见 Noisy, small peaks hidden	时间太长 Too long	4 倍扫描 → 2 倍 SNR 4× scans → 2× SNR
p1	越短带宽越宽 Shorter = wider bandwidth	越短功率越高 Shorter needs more power	短但功率不够：角度不足 Too short for the power: under-tipped	长：远处峰变弱甚至变负 Long: far peaks weak or negative	好的 5 mm 探头 10–15 μs Good 5 mm probe: 10–15 μs
TD (F1) 增量数	间接维分辨率 ↑ F1 resolution up	实验时间 ∝ 增量数 Time scales with increments	F1 方向峰被拉长 Peaks smeared along F1	时间太长 Too long	少增量可配 LP Few increments: add LP
SI / 补零	数字分辨率 ↑，谱线更平滑 Digital resolution up, smoother	没有新信息 No new information	谱线粗糙 Coarse lines	实际差别很小 Little practical difference	SI = TD 已相当于补零一次 SI = TD already doubles the points
LP	分辨率 ↑，省时间 Resolution up, saves time	加的是“假”数据 Adds “fake” data	— —	可能产生伪影 Can create artifacts	2D 少增量时效果最好 Best in 2D with few increments
LB	灵敏度 ↑ Sensitivity up	分辨率 ↓ Resolution down	噪声大 Noisy	峰变宽、裂分消失 Broad peaks, splitting lost	小于最尖峰线宽；^{1}H 0–0.5 Hz；^{13}C 5–15 Hz Below sharpest peak; ^{1}H 0–0.5 Hz; ^{13}C 5–15 Hz

碳谱去耦：zgdc 全程去耦，SNR 最高但有 NOE；zgig 只在 AQ 去耦，无 NOE，定量必须用它。 Carbon decoupling: zgdc decouples throughout (highest SNR, with NOE); zgig decouples only during AQ (no NOE), required for quantitative work.

必背公式、数字与因果链 Formulas, numbers and cause chains

公式（代入讲义面板的真实数值） Formulas, with the slide's real values

$\mathrm{DW} = 1/(2 \cdot \mathrm{SWH})$	$1/(2 \times 10000)$ = 50 µs	点间距 dwell time
$\mathrm{AQ} = \mathrm{TD} \times \mathrm{DW}$	65536×50 µs = 3.2768 s	真正接收的时间
$\mathrm{FIDRES} = 1/\mathrm{AQ}$	$1/3.2768$ = 0.305 Hz	FID 分辨率 resolution
$\mathrm{SW}_{\mathrm{ppm}} = \mathrm{SWH}/\mathrm{SFO1}$	$10000/500.29$ = 19.988 ppm	Hz 换算 ppm
$\mathrm{SFO1} = \mathrm{BF1} + \mathrm{O1}$	500.29 MHz + 2351.83 Hz	实际工作频率 actual frequency
$\mathrm{O1P} = \mathrm{O1}/\mathrm{BF1}$	$2351.83/500.29$ = 4.701 ppm	谱图中心 spectrum centre
$f_{\mathrm{Nyquist}} = \nu/2$	ν = 采样率 sampling rate	更高的频率会混叠 aliases above this
$M_z(t)/M_0 = 1 - e^{-t/T_1}$	$t = T_1$ 时 63%	T1 恢复 recovery
$\mathrm{SNR} \propto \sqrt{\mathrm{NS}}$	总扫描数 = NS × TD0	total scans
$\mathrm{d4} = 1/(4J)$	J = 145 Hz → 1.72 ms	HSQC 耦合延迟 coupling delay
线宽 $= 1/(\pi T_2)$	FID 衰减越快，峰越宽	faster decay = broader line

必背数字 Numbers to memorize

ADC 位数 bits	8 位 256 级；16 位 65 536 级
T1 恢复 recovery	1×：63% · 3×：95% · 5×：99.3% · 7×：99.9%
D1	1D：3–5 × T1 · 2D：1–3 × T1
定量 quantitation	常规误差可达 8%；定量条件 1%，耗时 5 倍
SW @ 500 MHz	^{1}H 8–10 kHz · ^{13}C 约 25 kHz
HSQC SW	^{1}H 约 20 ppm · ^{13}C 约 200 ppm
RG 上限 max	AVANCE III：203 · AVANCE NEO：101
p1	好的 5 mm 探头 10–15 µs
LB	^{1}H 0–0.5 Hz · ^{13}C 5–15 Hz
窗函数 window	HSQC：移位 sine² · COSY：不移位 sine²
钴 cobalt	化学位移范围约 20 000 ppm

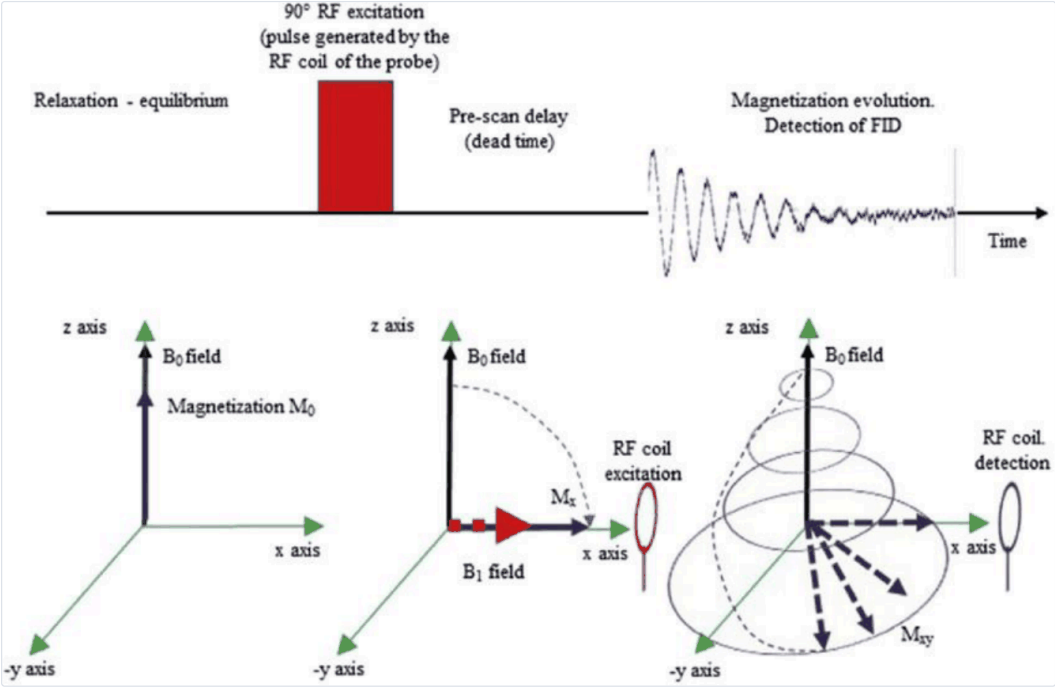
七条因果链 Seven cause chains

TD ↓ → AQ ↓ → FID 没衰减完就被切断 → 波纹 + 分辨率 ↓ + 定量 ↓	TD down → AQ down → FID truncated → wiggles, lower resolution, lost quantitation
RG ↑ → 信号超过 ADC 最高一级 → 削顶 → 畸变、假峰	RG up → signal passes the top ADC level → clipped → distortion, fake peaks
D1 ↓ → Mz 恢复不全 → 下一扫信号 ↓ → 长 T1 的峰被低估，定量出错	D1 down → Mz not recovered → less signal next scan → long-T1 peaks underestimated
p1 ↑ → 激发带宽 ↓ → 离 O1P 远的峰变弱或变负	p1 up → bandwidth down → peaks far from O1P weaken or go negative
Lock 差 → 磁场随时间漂移 → 线形差、2D 出现条纹	Poor lock → field drifts over time → poor lineshape, stripes in 2D
Shim 差 → 样品各处 B_0 不同 → FID 衰减快 → 峰宽 → 裂分信息丢失	Poor shims → B_0 varies across the sample → fast FID decay → broad peaks → splitting lost
LB ↑ → FID 尾部（噪声多）被压低 → SNR ↑，但线宽 ↑	LB up → noisy FID tail suppressed → SNR up, linewidth up

1.1 从线圈到放大器 From coil to pre-amplifier

讲义 p.2–3

我们真正测到的，是检测线圈里一个微伏级的振荡电流。
What we physically measure is a microvolt-level oscillating current in the detection coil.



脉冲 → 死时间 → 检测 FID；下方是对应的矢量图 讲义 p.2
Pulse → pre-scan delay (dead time) → FID detection, with the vector picture below

90° 射频脉冲把 z 轴上的净磁化 M_0 打到 xy 平面。

A 90° RF pulse from the probe's coil tips the net magnetization M_0 from z into the xy-plane.

磁化在 xy 平面进动、衰减，在线圈里感应出振荡电流，也就是 **FID**。

It precesses and decays in the xy-plane, inducing an oscillating current in the coil: the **FID**.

真实信号非常小，只有**微伏**量级。

The actual signal is very small, on the order of **microvolts**.

所以用**前置放大器**把信号放大，它通常就装在探头附近。

A **pre-amplifier**, normally found close to the probe, makes the signal larger.

放得近，是为了把电阻造成的信号损失降到最低；放大之后，电阻损耗就没那么要紧了。

This keeps signal lost to resistance to a minimum. Once amplified, resistive losses are less critical.

记忆钩子 Memory hook

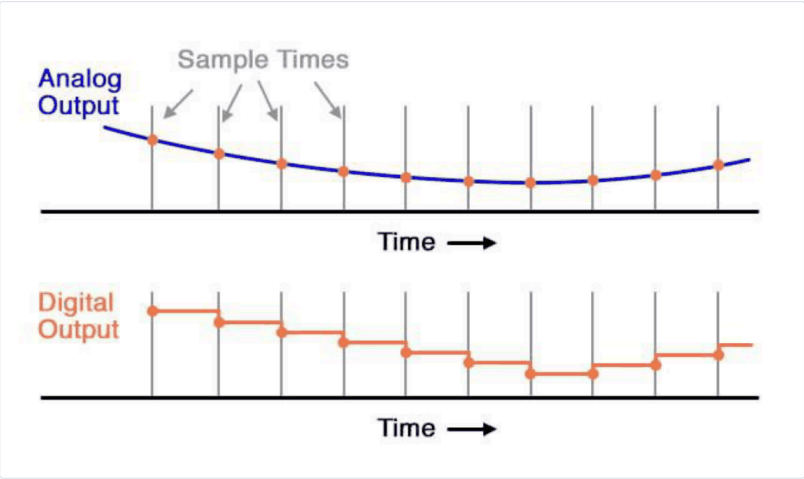
像录一段悄悄话：麦克风要贴着说话的人（前放贴着探头）。声音放大以后，线再长也不怕。

Like recording a whisper: the mic goes right next to the speaker. Once amplified, a long cable hardly matters.

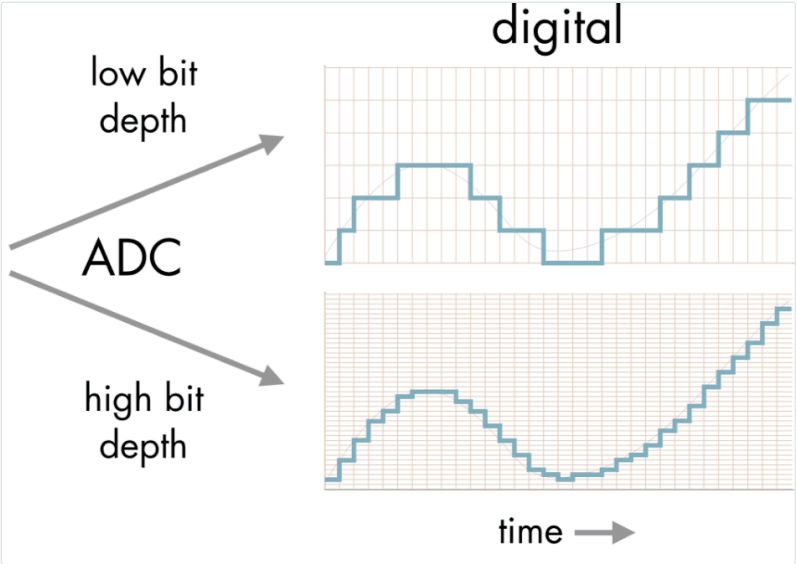
1.2 ADC：把模拟信号变成数字 The analogue-to-digital converter

讲义 p.4–5

ADC 做两次离散化：时间上按固定间隔取点，幅度上只能落在有限的台阶。
The ADC discretizes twice: in time (a fixed sampling interval) and in amplitude (a limited number of levels).



按固定间隔取点，值近似为最近的数字量，所以有误差 讲义 p.4
Sampled at fixed times and approximated, hence some error



位数低：阶梯粗。位数高：贴近原曲线 讲义 p.5
Low bit depth is coarse; high bit depth follows the curve

产生 FID 的振荡是**模拟**信号，必须数字化才能处理；这由**模数转换器 ADC** 完成。

The oscillation that gives rise to the FID is **analogue** and must be digitized for processing, using an **analogue-to-digital converter**.

ADC 每隔固定时间采样一次，并近似该点的值。

The ADC samples at a given interval and approximates the value at that point.

所以模拟信号和数字信号之间总有一些**误差**。

There is some **error** between the analogue and digitized signal.

模拟信号连续，可取无穷多个值；ADC 只能给离散值，受它的**分辨率**限制。

An analogue signal is continuous with infinitely many values. The ADC assigns a discrete value limited by its **resolution**.

8 位 ADC 有 256 级，16 位有 65 536 级。

An 8-bit ADC has 256 levels, while a 16-bit has 65 536.

分辨率越高 = 数字化越准确。

Higher resolution = more accurate digitization.

记忆钩子 Memory hook

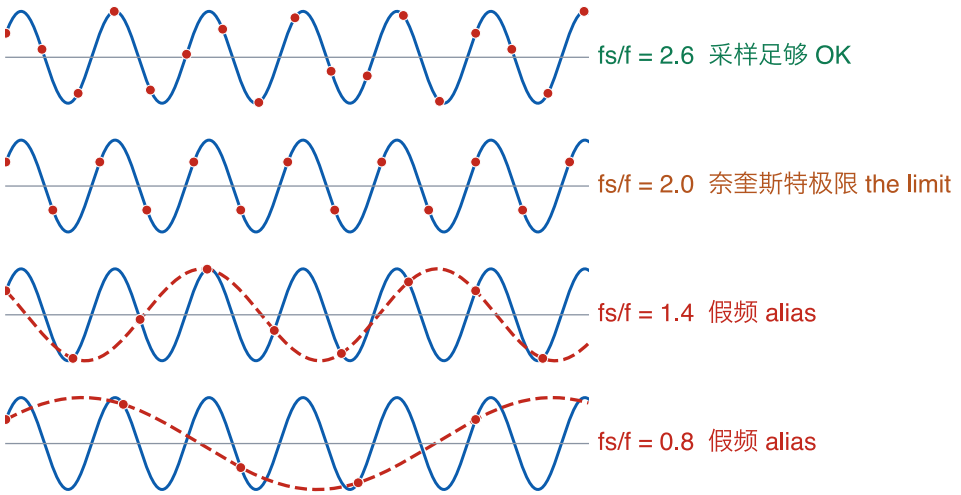
n 位 → 2^n 级。想象楼梯：台阶越细，越贴近真实的斜坡。

n bits → 2^n levels. Like stairs: finer steps follow the true slope more closely.

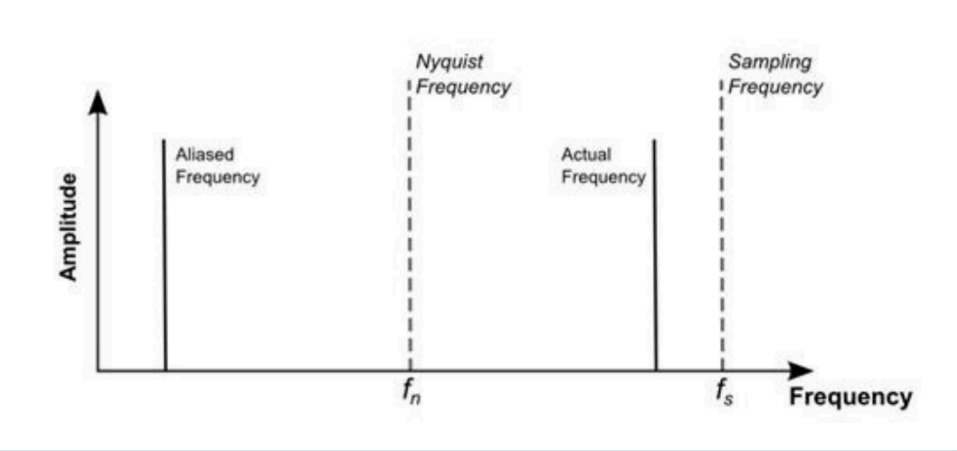
1.3 采样率与混叠 Sampling rate and aliasing

讲义 p.6–7

采样率至少是最高频率的两倍；否则高频信号会伪装成并不存在的低频信号。
Sample at least twice as fast as the highest frequency, or high frequencies masquerade as low ones that are not there.



蓝线是真实信号，红点是采样点。红虚线：只看采样点时“看到”的假信号 模拟 simulation
Blue: true signal. Red dots: samples. Red line: the fake signal the samples imply



真实频率超过 f_n 后，会以 $f_s - f$ 的假频率出现 讲义 p.7
An actual frequency above f_n appears at the aliased frequency $f_s - f$

$$f_{\text{Nyquist}} = \frac{1}{2}\nu$$

快速采样其实很难。
Turns out sampling fast is hard.

必须以最高频率**两倍**的速度采样，才能拿到全部正确信息。
We need to sample **twice as fast** as the highest frequency to get all the correct information.

奈奎斯特频率：给定采样率 ν 时，能正确表示的最高频率。
The **Nyquist frequency** is the highest frequency that can be represented at a given sample rate ν.

采样太慢就会出现“假”信号，叫**混叠**（aliasing）。
If we sample too slowly we get “fake” signals, called **aliasing**.

采样率不是直接设的，而是由采集参数**间接**控制（见 2.1）。
The sampling rate is controlled **indirectly** through acquisition parameters (see 2.1).

记忆钩子 Memory hook
电影里的车轮：转得太快时，看起来像在慢慢倒转。摄像机的“采样率”不够，大脑就看到了假运动。
The wagon-wheel effect: a fast wheel on film seems to turn slowly backwards. The camera’s sampling rate is too low.

2.0 一张面板看懂全部基本参数 The basic parameter panel

讲义 p.8–9

这就是一次简单 1D 氢谱的全部设置。接下来逐个讲：设错时谱图长什么样。

These are the most basic parameters of a simple 1D proton experiment. Next: what spectra look like when you set each one wrong.

参数	讲义里的值	中文	Meaning	节
PULPROG	zg	脉冲程序	Current pulse program	—
TD	65536	FID 点数	Size of FID	2.1
SWH	10000 Hz, 19.9883 ppm	谱宽	Sweep width in acquisition direction	2.4
AQ	3.2768 s	采集时间	Acquisition time	2.1
RG	0.25	接收增益	Receiver gain	2.5
DW	50 µs	驻留时间	Dwell time	2.1
DE	6.50 µs	预扫描延迟	Pre-scan delay	2.7
D1	30 s	弛豫延迟	Relaxation delay; 1–5 × T1	2.7
DS	0	空扫数	Number of dummy scans	2.9
NS	16	扫描数	Number of scans	2.9
TD0	1	外层循环数	Number of averages in 1D	2.9
SFO1	500.2923518 MHz	通道 1 频率	Frequency of channel 1	2.10
O1	2351.83 Hz, 4.701 ppm	频率偏置	Offset of channel 1	2.10
NUC1	^{1}H	观测核	Nucleus for channel 1	—
P1	14.39 µs	90° 脉冲长度	F1 channel: high power pulse	2.12
PLW1	10.854 W, –10.36 dB	脉冲功率	F1 channel: power level for pulse	2.13

算一遍 Worked example

这些数互相锁死：

DW = 1/(2×10000) = 50 µs

AQ = 65536×50 µs = 3.2768 s

SW = 10000/500.29 = 19.988 ppm

O1P = 2351.83/500.29 = 4.701 ppm

–10·lg(10.854) = –10.36 dB

The values are locked together: you choose TD and SW, and the software computes DW and AQ. Power in dB is $-10 \cdot \log_{10}$ of the power in watts, so a larger number in dB means lower power.

采集软件一般跟着谱仪厂商走。

Acquisition software typically depends on the vendor.

Bruker 是最大的 NMR 厂商，采集和处理都用 **TopSpin**；学术用户可在 Bruker 官网免费下载（用于处理）。

Bruker, the leading manufacturer, uses **TopSpin** for both; free download for academic users (processing).

VnmrJ 是 Varian 谱仪的软件，Varian 停业后已开源。

VnmrJ was for Varian spectrometers before they went out of business (now open source).

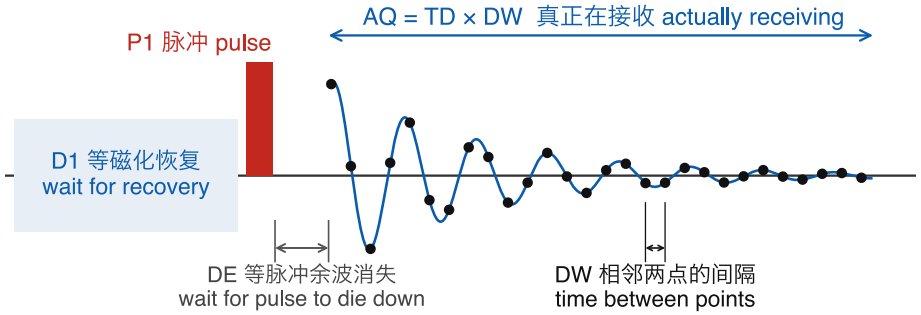
Magritek、Oxford Instruments 等小厂商有自己的软件。

Smaller vendors (Magritek, Oxford Instruments) have their own software.

2.1 TD、SW、DW、AQ 四者的关系 How TD, SW, DW and AQ connect

讲义 p.10, 19, 28

你只设 TD 和 SW；DW 和 AQ 是算出来的。分辨率取决于“听了多久”，也就是 AQ。
You set TD and SW; DW and AQ follow. Resolution depends on how long you listen, which is AQ.



一次 scan 总时间 ≈ D1 + P1 + DE + AQ，重复 NS 次　One scan ≈ D1 + P1 + DE + AQ, repeated NS times

一次扫描的时间轴：黑点就是 TD 个采样点 模拟 simulation
One scan’s timeline: the black dots are the TD sampled points

讲义里的三组真实数据 Three real cases from the slides

来源 Source	TD	SWH (Hz)	AQ (s)	1/AQ (Hz)
p.9 500 MHz ^{1}H	65536	10000	3.277	0.305
p.14 80 MHz ^{1}H	16384	1612.9	5.079	0.197
p.51 HSQC, F2	2048	7142.9	0.143	6.975

$$\mathrm{AQ} = \mathrm{TD} \times \mathrm{DW} = \frac{\mathrm{TD}}{2 \cdot \mathrm{SWH}}$$

TD：生成谱图所采的点数。它决定谱图分辨率，但必须同时考虑 SW。

TD is the number of points sampled to create the spectrum. It determines the resolution, but you must also consider the sweep width.

SW：扫描宽度，基本上就是谱图有多宽。

SW is the sweep width: basically how wide the spectrum is.

DW：FID 中相邻两点之间的时间。

DW (dwell time) is the time between points in the FID.

AQ：真正在接收信号的时间，不是一次扫描的总时间。它由 TD 和 SW 算出，一般不直接设置。

AQ is not the total time for a scan but how long we are actually receiving. Calculated from TD and SW; typically not set directly.

所以 TD 不变而 SW 加宽 → DW 变小 → AQ 变短 → 分辨率变差。

So widening SW at fixed TD shortens DW and AQ, and resolution gets worse.

记忆钩子 Memory hook

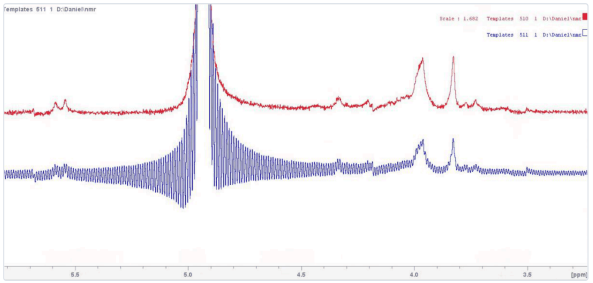
连拍照片：TD 是张数，DW 是间隔，AQ 是总时长。想看清缓慢的变化（细小的裂分），就要拍得够久。

Burst photos: TD is the count, DW the interval, AQ the total time. To catch slow changes (fine splitting) you must shoot for long enough.

2.2 TD 太小：谱图上的三种症状 TD too low: three looks in the spectrum

讲义 p.11–13

TD 太小 → FID 还没衰减完就被切断（截断）。症状从一眼能看出到非常隐蔽。
Too low a TD truncates the FID. The symptoms range from obvious to subtle.

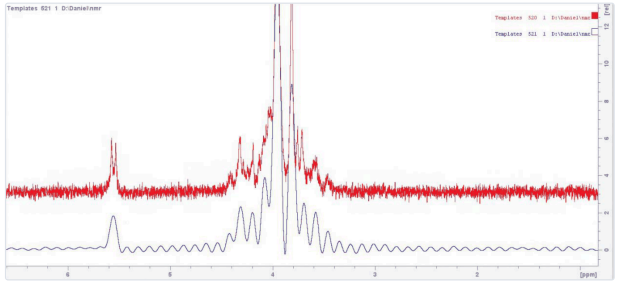


① 波纹
Waves

讲义 p.11

蓝线在强峰两侧出现波浪：这就是**截断伪影**。

Truncation artifacts look like **waves** in the spectrum (blue trace).

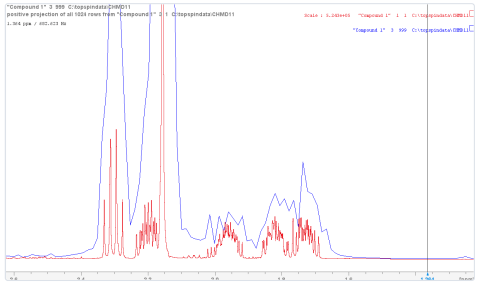


② 波纹 + 分辨率下降
Waves plus lost resolution

讲义 p.12

同样有截断伪影，但还多了一个问题：**分辨率也变差了**。

Truncation artifacts again, but **loss of resolution** also shows.



③ 只有分辨率下降
Lost resolution only

讲义 p.13

最隐蔽：分辨率大幅损失，却**没有典型的波纹**。蓝线把红线的细节全糊掉了。

More subtle: a large loss in resolution, but the common truncation artifacts **aren't there**.

出错 What goes wrong

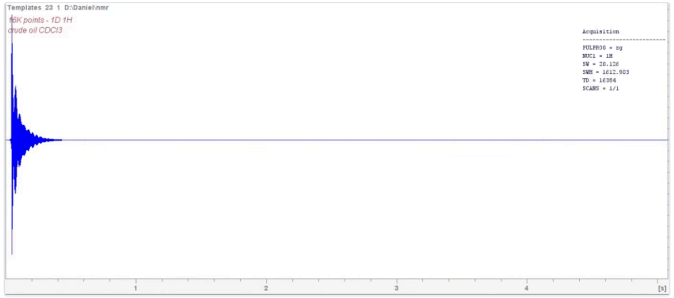
没有波纹 ≠ TD 够用。别只看谱图，要回去看 FID（下一页）。

No wiggles does not mean TD is sufficient. Don't judge from the spectrum alone: go back and look at the FID (next page).

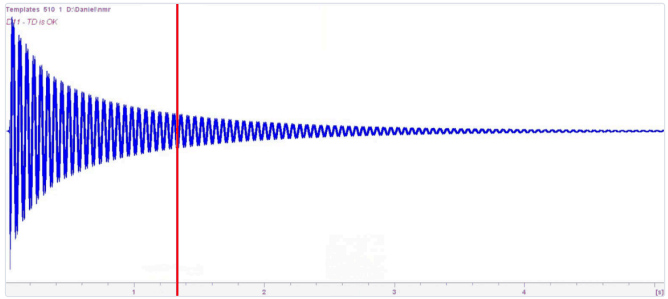
2.3 判断 TD 够不够：看 FID 的尾巴 Is TD enough? Look at the FID's tail

讲义 p.14–15

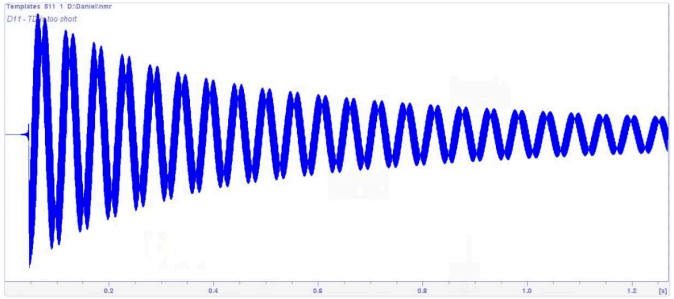
AQ 结束前，FID 应该已经衰减进噪声。还在振荡就是截断，定量也不可信。
By the end of AQ the FID should have decayed into the noise. If it is still ringing, it is truncated and quantitation cannot be trusted.



原油，16K 点：很早就衰减干净 讲义 p.14
Crude oil, 16K points: decays well



TD 合适：约 4–5 s 时才衰减完 讲义 p.15
TD is OK: fully decayed by about 4–5 s



同一样品，AQ 只有 1.2 s：结束时还在强烈振荡 讲义 p.14–15
Same sample, AQ only 1.2 s: still ringing strongly at the end

截断还会造成**定量信息丢失**。

Truncation leads to **loss of quantitative information**.

第一个 FID 衰减得很好。那第三个呢?

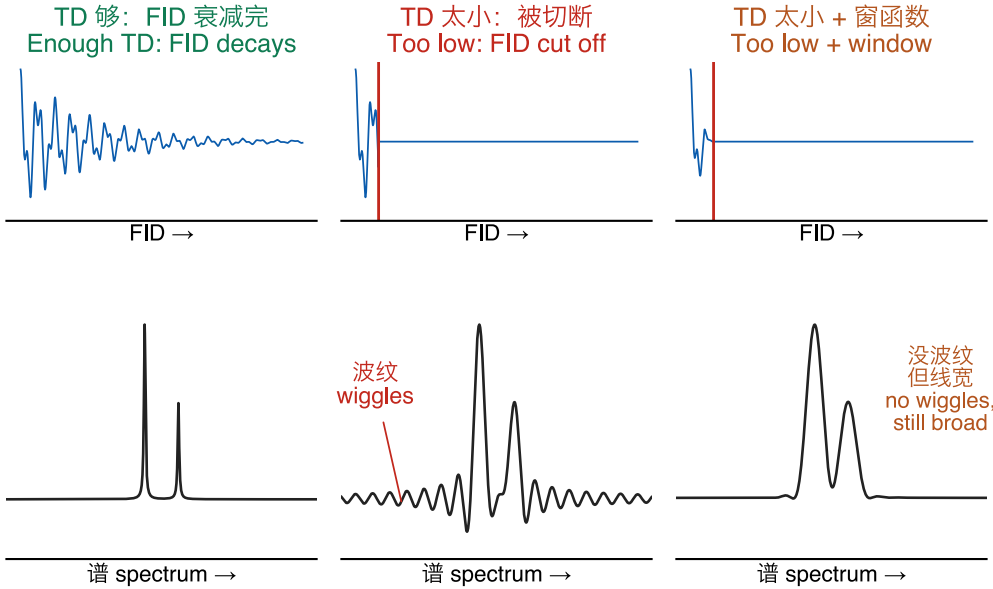
Here the top decays pretty well. What about the bottom?

TD 更高（采集更久）时，FID 能充分衰减。

A higher TD (longer acquisition) lets the FID decay.

AQ 只有 **1.2 s** 时，FID 被明显截断，带来伪影和定量数据丢失。

A **1.2 s** acquisition time considerably truncates the FID, giving artifacts and loss of quantitative data.



机理：同一信号的三种命运 模拟 simulation
The mechanism: one signal, three fates

讲义外补充 Beyond the slides

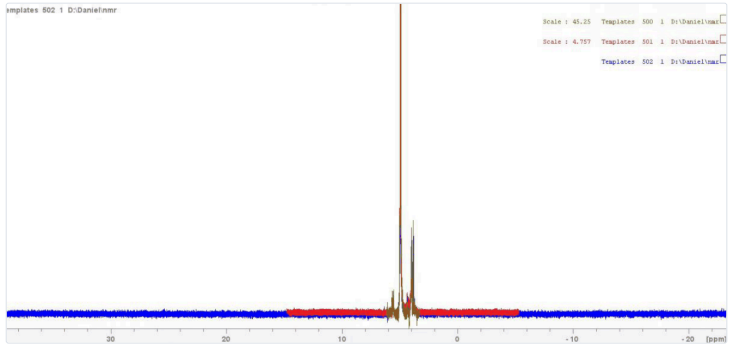
FID 被突然切断，等于乘了一个矩形窗；FT 后每条谱线都带上 sinc 波纹。窗函数能磨平波纹，但线宽回不去。症状 ③ 很可能就是这样来的。

Cutting the FID multiplies it by a rectangle, so each line gets sinc wiggles after FT. A window smooths the wiggles but cannot restore the linewidth: a likely origin of symptom ③.

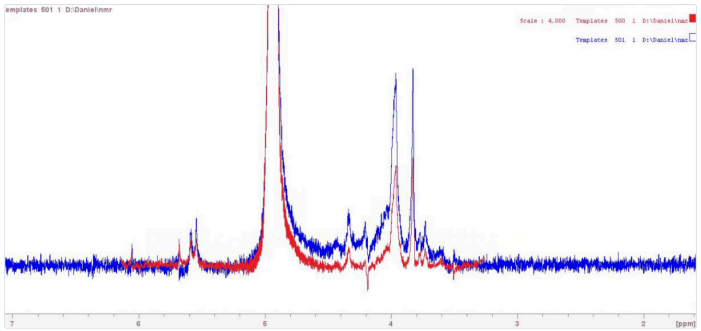
2.4 SW：谱宽要盖住所有信号 SW must cover every signal

讲义 p.16–18

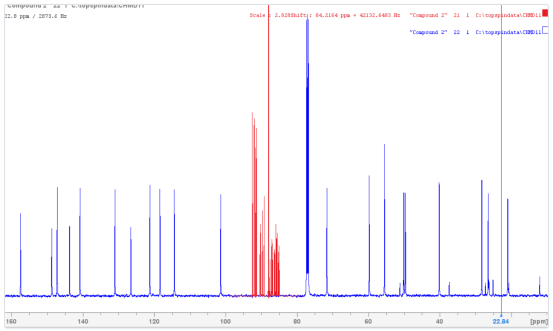
SW 就是谱图有多宽。它必须盖住所有有信号的区域，而不同核的范围差很多。
SW is how wide the spectrum is. It must cover every region with signal, and the range differs greatly between nuclei.



不同 SW 叠在一起：信号都在中间一小段 讲义 p.16
Different SWs overlaid: the signal sits in a small central region



放大看：窄窗似乎也够用，但这个样品没有芳香峰 讲义 p.17
Zoomed in: even a narrow window seems fine, but this sample has no aromatics



^{13}C（蓝）比 ^{1}H（红）宽得多 讲义 p.18
^{13}C (blue) spans far more than ^{1}H (red)

要确保 SW 盖住所有有信号的区域。	We want to make sure that we cover all of the regions that have signal.
例：这个样品在所关心的带宽之外好像没有信号，连最窄的选项都够用……	Here there seem to be no signals outside the bandwidth we care about, so even the smallest option would be fine…
……但是芳香峰会出现在哪?	…however, where would aromatics resonate?
碳谱的范围比氢谱宽得多，所以需要更宽的 SW。	Carbon spans a much wider range and thus requires a wider SW.
500 MHz 上常见值：**氢谱约 8–10 kHz，碳谱约 25 kHz。**	On a 500, a common proton spectrum may be **~8–10 kHz, while carbon would be ~25 kHz**.

出错 What goes wrong

看一个样品的谱图就断定窄窗够用，很危险：只是这个样品刚好没有落在窗外的峰。	Judging SW from one sample is risky: this sample just happens to have no peaks outside the window.

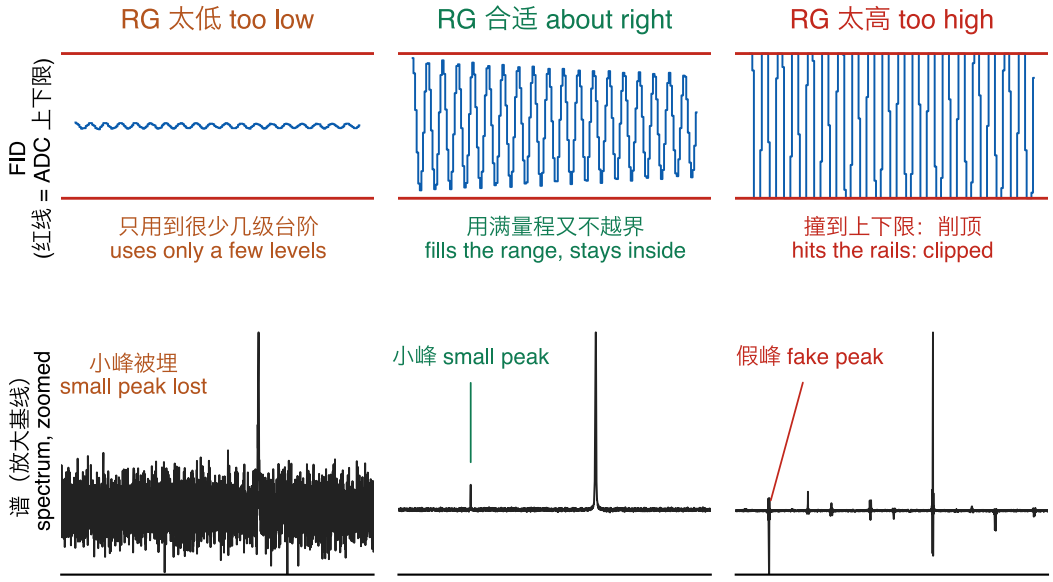
讲义外补充 Beyond the slides

换算一下：500 MHz 谱仪上碳的频率约 125.8 MHz，所以 25 kHz ≈ 200 ppm；氢的 10 kHz ≈ 20 ppm。按 Hz 只差 2.5 倍，按 ppm 差 10 倍。	On a 500 the carbon frequency is about 125.8 MHz, so 25 kHz ≈ 200 ppm, while proton 10 kHz ≈ 20 ppm: 2.5× in Hz, but 10× in ppm.

2.5 RG：接收增益如何工作 How receiver gain works

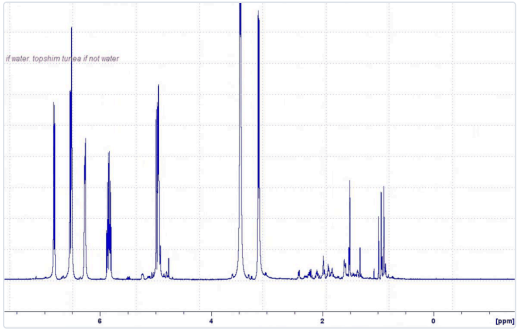
讲义 p.20–23

RG 把所有峰一起放大：小峰更好读，但一旦超过 ADC 最高一级，就被削顶（溢出）。
RG scales all peaks up together: small peaks read better, but pass the top ADC level and the signal is clipped (overflow).



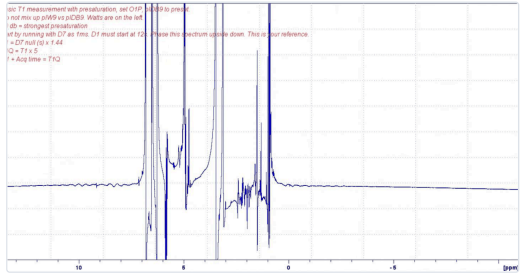
同一个“一强一弱”样品，三种 RG 模拟 simulation

One sample with a strong and a weak peak, at three RG settings



RG 合适：谱图没有明显问题 p.20

Fine gain: no immediate issues



RG 过高：讲义说的 “nasty spectra” p.21

Too high: “nasty spectra”

RG 是接收增益，基本上就是接收器有多灵敏。设对它是好谱图的关键。

调高：小峰的信号更好。

太高：ADC “溢出”，谱图很难看。

合适的值通常取决于样品的**强度和动态范围**。

机制：ADC 把连续信号变成离散台阶。希望峰占用多级台阶，才分得清强弱。RG 相当于把所有峰等比放大，超过最高一级就削顶。

RG is the receiver gain: essentially how sensitive the receiver is. Setting it properly is critical to good quality spectra.

Setting it higher gives better signal from small peaks.

If it is too high, the ADC will “overflow”, giving nasty spectra.

This typically depends on the **intensity and dynamic range** of your sample.

Mechanism: the ADC converts the signal into discrete levels. Peaks should use multiple levels so we can tell them apart in intensity. RG scales all peaks up; pass the highest level and you clip the peak.

讲义勘误 Slide correction

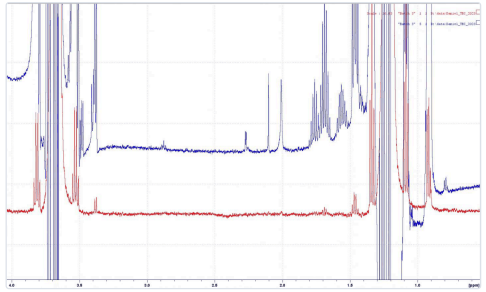
p.23 写的 “continuous digital signal” 是笔误，应为 continuous **analogue** signal。

p.23 says “continuous digital signal”; it should read continuous **analogue** signal.

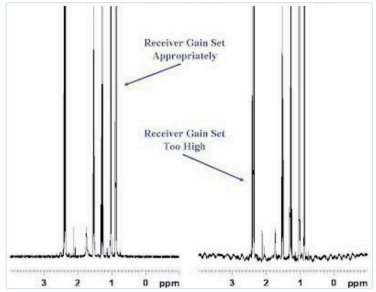
2.6 RG：太低、太高和怎么设 RG too low, too high, and how to set it

讲义 p.22, 24–27

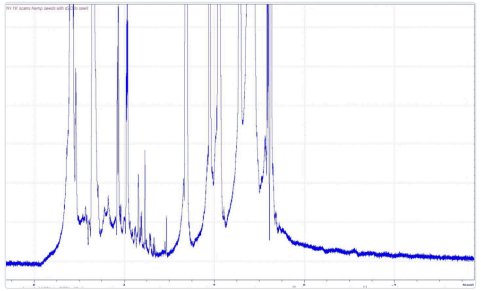
RG 低：不失真但丢灵敏度。RG 高：失真。做法：单扫下调到畸变消失，或直接用 rga。
RG too low: no distortion, lost sensitivity. Too high: distortion. Lower it on a single scan until distortions go away, or use rga.



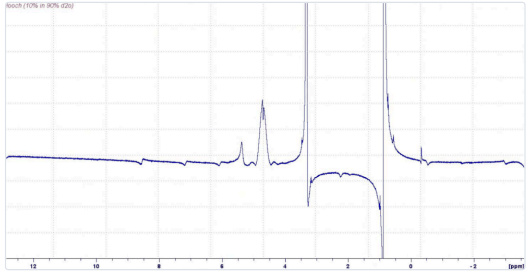
太低：RG 0.25（红）vs 101（蓝） p.25
Too low: 0.25 (red) vs 101 (blue)



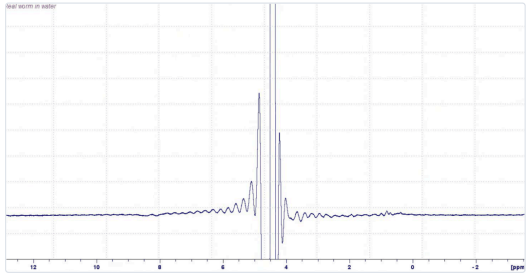
单扫氢谱：左 RG 合适，右过高 p.27
Single scan: appropriate (left), too high (right)



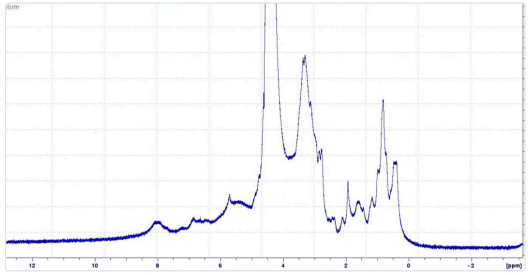
太高但隐蔽：从基线看增益略高 p.26
Subtly too high: judged from the baseline



太高：基线异常，有时还像真峰 p.26
Too high: odd baseline, can look like real peaks



活蚯蚓在水中，不压水：RG 没设好 p.22
Living earthworm in water, no suppression



同一样品压水后：所有峰都正常可见 p.22
With solvent suppression: all peaks visible

太低：谱图多半不会失真，却可能白白丢掉相当多的 SNR。对最小的峰尤其致命！

Too low: probably no distortion, but potentially throwing away a considerable amount of SNR. Especially critical for the smallest peaks!

太高：效果也可能很隐蔽。从基线能看出增益略高；有时看起来还像真实的峰。

Too high: effects can be subtle. The baseline suggests the gain is a bit too high; it can sometimes look like real peaks.

绝对值取决于谱仪类型：AVANCE III（有用的）上限 **203**，AVANCE NEO 上限 **101**。

The **absolute value** depends on the spectrometer: on an AVANCE III the (useful) max is **203**, on AVANCE NEO it is **101**.

凡是涉及碳的实验，通常都用最大值。

For anything involving carbon you typically want the max value.

手动设：跑单扫氢谱，把 RG 往下调，直到畸变消失。

By hand: run a single-scan proton and lower RG until distortions go away.

自动设：Bruker 上的程序 **rga**。

Automatically: the **rga** program on Bruker.

记忆钩子 Memory hook
录音的音量旋钮：调小了细节听不清，但不会破音；调大了会破音，而且事后修不回来。

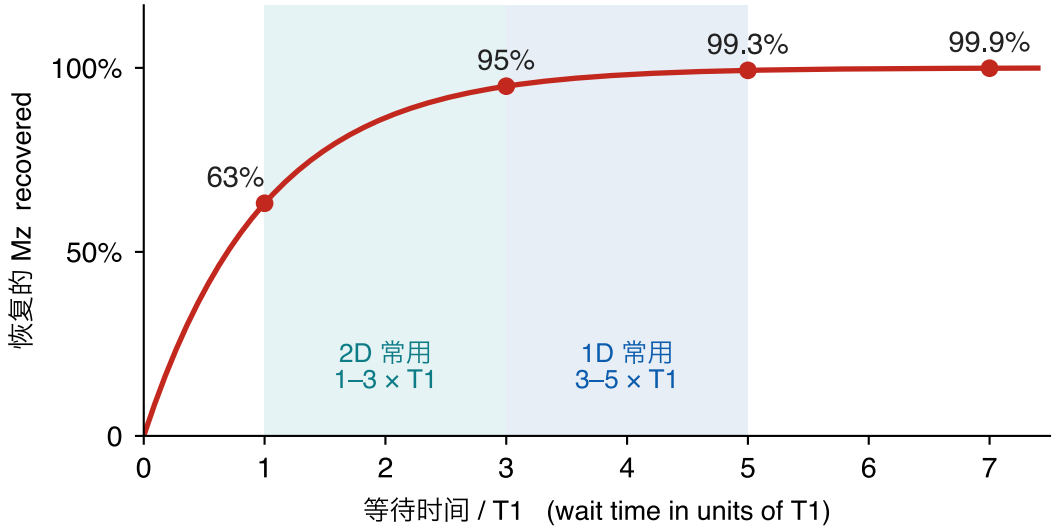
A recording level knob: too low and detail is lost, but nothing breaks up; too high and it distorts.

2.7 DE、D1 和 T1 DE, D1 and T1

讲义 p.28–32

两次扫描之间要等磁化回到 z 轴。T1 决定需要等多久，D1 是你实际等了多久。

Between scans, magnetization must return to the z-axis. T1 sets how long that takes; D1 is how long you actually wait.



等待时间与恢复比例 模拟 simulation

Recovery versus wait time

记忆钩子 Memory hook

口诀 1-3-5 对应 63-95-99：等 1 个 T1 回来 63%，3 个回来 95%，5 个回来 99%。

1-3-5 goes with 63-95-99: wait 1, 3 or 5 T1 and you get back 63%, 95% or 99%.

讲义勘误 Slide correction

p.32 写 7 × T1 约 99.991%。按 $1 - e^{-7}$ 算应为 **99.91%**。如果考试直接考这个数，答 99.9% 并注明公式最稳妥。

p.32 gives ~99.991% at 7 × T1, but $1 - e^{-7}$ = **99.91%**. If tested, answer 99.9% and show the formula.

延迟对复杂实验的工作方式至关重要，基础延迟同样重要。

Delays are critical to how more complex experiments work, and the basic ones are also important.

DE（预扫描延迟）：通常用同一个线圈发射和接收，所以要等脉冲余波消失，免得把它当成信号。

DE (pre-scan delay): it is very common to use the same coil to pulse and receive, so we wait for the pulse to die down to avoid seeing it as our signal.

D1（弛豫 / 循环延迟）：开始下一次扫描前等多久。对定量工作至关重要。

D1 (relaxation or recycle delay): how much you wait before the next scan. Critical for quantitative work.

T1：每个自旋都有自己的 T1，表示约 **63%** 的磁化恢复到 z 轴所需的时间。

T1: each spin has a specific value, telling us how long it takes for ~**63%** of magnetization to recover to the z-axis.

1D：D1 常设为 3–5 × T1。**2D**：通常更低，约 1–3 × T1。

1D: D1 is commonly set to 3–5 × T1. **2D**: usually lower, ~1–3 × T1.

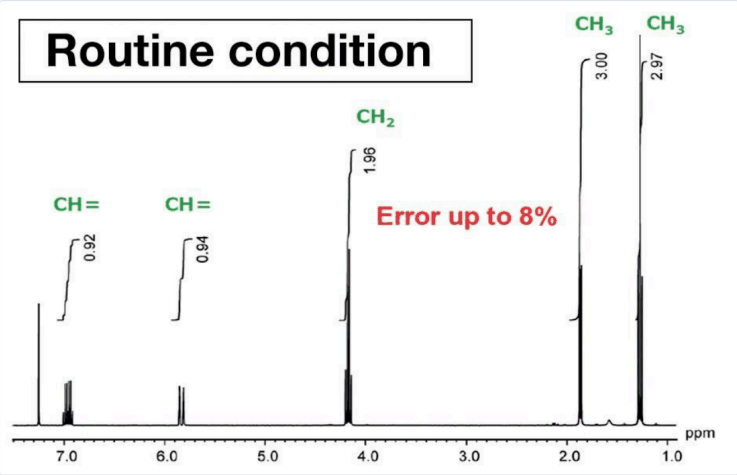
3 × T1 回来约 95%；5 × T1 约 99.3%；7 × T1 约 99.9%。

After 3 × T1, ~95% is back; after 5 × T1, ~99.3%; after 7 × T1, ~99.9%.

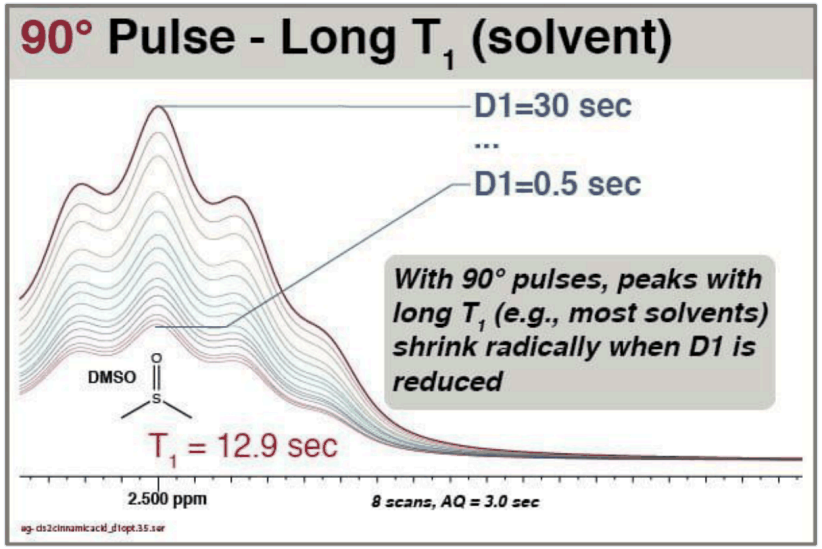
2.8 D1 的取舍 The D1 trade-off

讲义 p.32–33

D1 越长越准，但越慢。要不要等，取决于你要的是结构还是定量。
Longer D1 is more accurate but slower. Whether to wait depends on whether you need structure or quantitation.



常规条件下，积分误差可达 8%：谱图看起来却完全正常 讲义 p.32
Routine conditions: integration error up to 8%, though the spectrum looks perfectly normal



DMSO，T1 = 12.9 s：D1 从 30 s 降到 0.5 s，峰急剧缩小 讲义 p.33
DMSO, T1 = 12.9 s: peaks shrink radically as D1 drops from 30 s to 0.5 s

为什么不总用长 D1? 为了更准的定量信息而多花时间，并不总是值得。

做结构解析时，常规条件就够了。

对同一分子做定量研究，误差降到 **1%**，但花了 **5 倍**时间！

D1 太短：磁化恢复不全，下一次扫描的强度就达不到那么高。影响可能相当大。

用 90° 脉冲时，长 T1 的峰（如多数溶剂）在 D1 缩短后急剧变小。

但 D1 短意味着同样时间能多扫几次，所以这个取舍并不简单！

Why not always use a long D1? It is not always worth the additional time to get more quantitative information.

For structure elucidation this is fine.

A quantitative study of this same molecule reduced error to **1%**, however it took **5 times** as long!

D1 too short: magnetization does not fully recover, so the next scan will not reach as high an intensity. This can be a considerable issue.

With 90° pulses, peaks with long T1 (e.g., most solvents) shrink radically when D1 is reduced.

However, lower D1 means more scans, so the trade-off is not super simple!

出错 What goes wrong

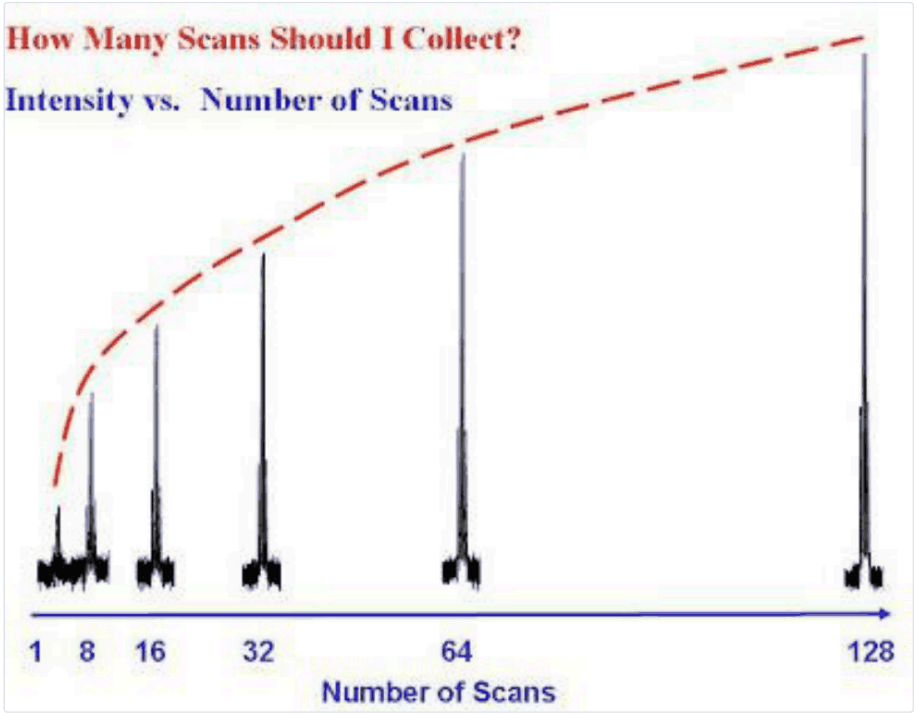
D1 不够时，谱图不会报警：峰形完美，只是积分是错的。

A short D1 raises no alarm: the peaks look perfect, only the integrals are wrong.

2.9 NS、DS、TD0：扫描数 Number of scans

讲义 p.34

SNR 只按扫描数的平方根增长：想把 SNR 翻倍，要花 4 倍时间。
SNR grows only with the square root of the number of scans: doubling SNR costs four times the time.



1 到 128 次扫描：强度上升，噪声相对下降，而且越往后越慢 讲义 p.34
From 1 to 128 scans: signal rises above the noise, with diminishing returns

NS	1	4	16	64	256
SNR	×1	×2	×4	×8	×16
时间 time	×1	×4	×16	×64	×256

$$\mathrm{SNR} \propto \sqrt{\mathrm{NS}}$$

NMR 里每一次采集叫一次**扫描**（scan）。
Each acquisition in NMR is called a **scan**.

通常要多次扫描才有好信号，所以要提高 **NS**。
Typically we need multiple scans to get good signal and must raise **NS**.

DS（空扫）：和正常扫描做完全一样的事，只是把数据丢掉。
DS (dummy scans) essentially do the same thing as normal scans, but discard the data.

TD0：把扫描数再乘一次，通常效果等同于提高 NS。总扫描数 = NS × TD0。
TD0 multiplies the number of scans and is generally the same as raising NS. Total scans = NS × TD0.

SNR 随扫描数的**平方根**增长。
The SNR scales with the **square root** of the number of scans.

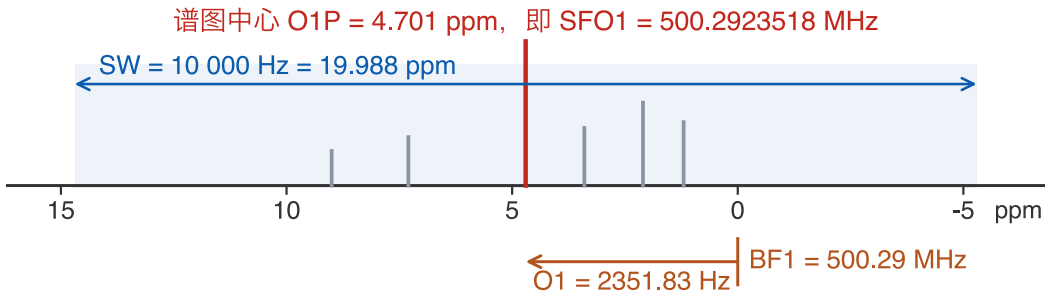
讲义外补充 Beyond the slides

为什么要空扫？前几次扫描时体系还没进入稳态，信号强度不一样。先空扫几次，让后面每次扫描都一样。
Why dummy scans? The first few scans have not reached a steady state, so their intensities differ. Dummy scans make the recorded ones consistent.

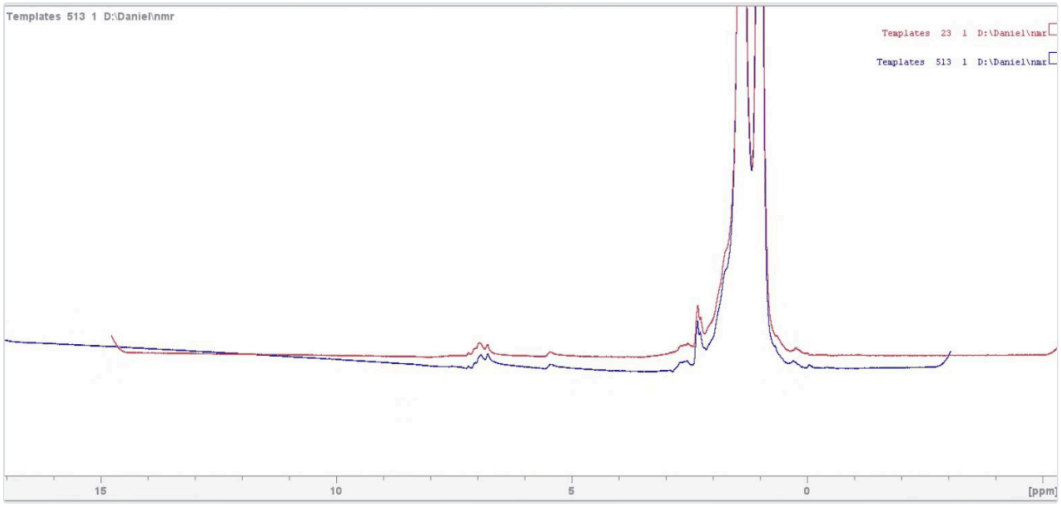
2.10 SFO1、O1、O1P：谱图中心 SFO1, O1, O1P: the spectrum's centre

讲义 p.35–37

这三个数说的是同一件事：谱图中心在哪。SFO1 是绝对频率，O1 用 Hz，O1P 用 ppm。
All three say one thing: where the centre of the spectrum is. SFO1 is absolute, O1 is in Hz, O1P is in ppm.



用讲义的真实数值画的频率轴 模拟 simulation
The frequency axis, drawn with the slide's real values



O1P 不同，数据基本一样，只是向两边延伸的距离不同 讲义 p.37
Different O1P: pretty much the same data, extending different distances each way

$$\mathrm{SFO1} = \mathrm{BF1} + \mathrm{O1}$$

SFO1：脉冲和接收实际发生的频率。

SFO1 is the actual frequency that pulsing and receiving occur at.

“500 MHz” 谱仪并不正好是 500 MHz；此例**基频**是 500.29 MHz。

In general, a 500 MHz spectrometer is not exactly at 500 MHz; here the **base frequency** is 500.29.

绝对频率用起来很麻烦，所以通常用相对值。

Using the absolute frequency is cumbersome, so relative values are more common.

O1P：谱图中心的 ppm 值。

O1P is where the centre of your spectrum is, in ppm.

O1：同一个值，只是单位是 Hz。

O1 is the same value, but in Hz.

它们三个关系非常紧密，改一个，另外两个跟着变。

SFO1, O1 and O1P are all very closely related.

记忆钩子 Memory hook
手电筒：O1P 是对准哪里，SW 是光圈开多大。

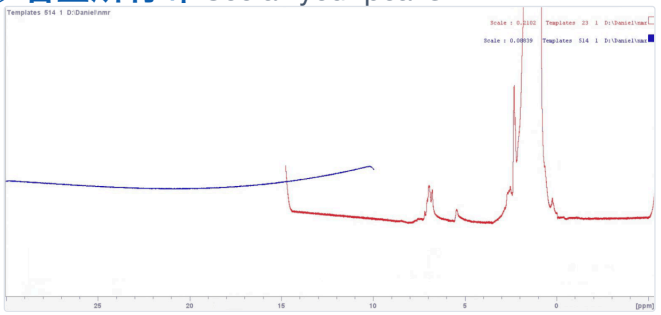
A flashlight: O1P is where you point it, SW is how wide the beam is.

2.11 O1P 设错的三个后果 Three reasons O1P matters

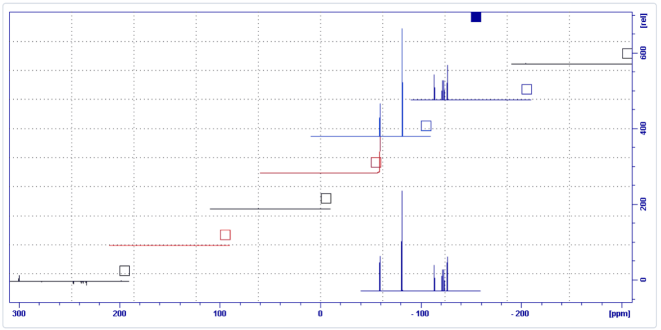
讲义 p.38–41

① 看不到峰，② 激发不均匀，③ 溶剂压不掉。
① You miss peaks. ② Excitation is uneven. ③ The solvent is not suppressed.

① 看全所有峰 See all your peaks



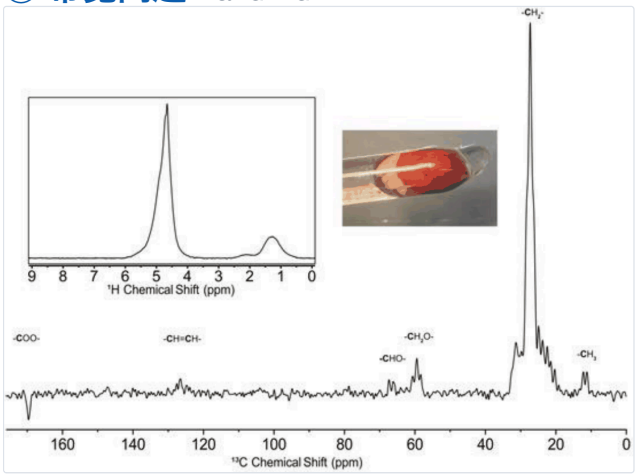
O1P 太偏：蓝谱一个峰也没看到 p.38
Blue misses all the peaks



^{19}F 分段“扫描”：只有两段看到峰 p.39
A ^{19}F “scan” across frequencies

氢谱少见，但金属核完全可能：**钴的范围约 20 000 ppm**！范围很宽又不知道位置时，可能要做多次实验。把 O1P 放在你关心的峰的中心。
Uncommon for proton, but it can happen for metals: **cobalt spans ~20 000 ppm**! With a very wide range and unknown shifts you may need multiple experiments. Choose O1P in the centre of the peaks you care about.

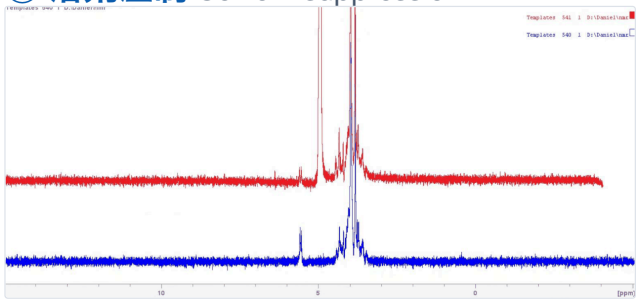
② 带宽问题 Bandwidth



^{13}C 谱：约 170 ppm 处的 COO 峰几乎消失还变负 p.40
The ~170 ppm COO peak nearly vanishes and goes negative

脉冲本应是完美方形，对所有 ppm 作用相同，但实际并非总是如此。这通常是**杂核**的问题。此例探头性能差，170 ppm 附近的峰实际受到约 **180° 脉冲**！
Pulses are expected to be perfectly square (affecting all ppm shifts the same), but this is not always the case. Usually problematic for **heteronuclei**. Here, due to poor probe performance, the peak at ~170 ppm experiences a **~180° pulse**!

③ 溶剂压制 Solvent suppression



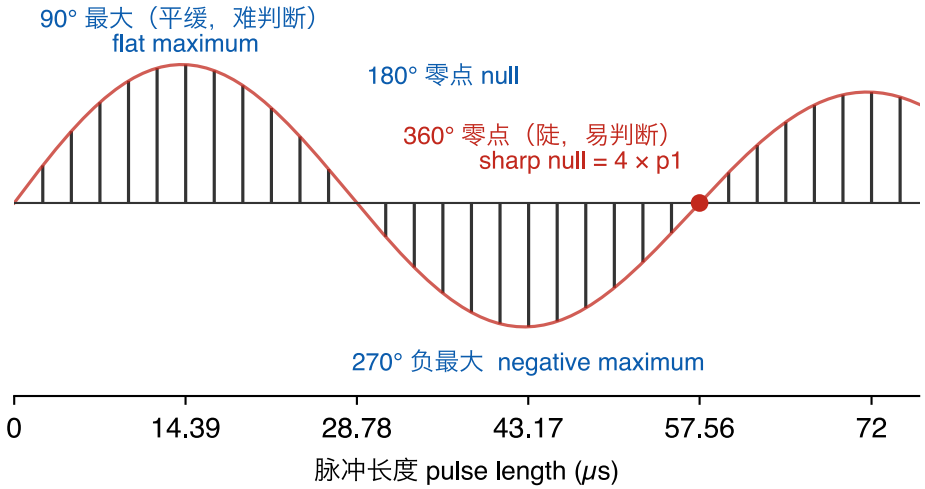
O1P 设错：水峰没被压掉 p.41
Wrong O1P: water not suppressed

最常见的压溶剂方法**只对 O1P 处起作用**，所以 O1P 必须设在要压的溶剂共振上。设错了水就压不掉，有时还会带来畸变。
Most common solvent suppression approaches **only work on the O1P**, so it must be set to the solvent resonance. Set incorrectly, the water is not suppressed, sometimes with distortions as well.

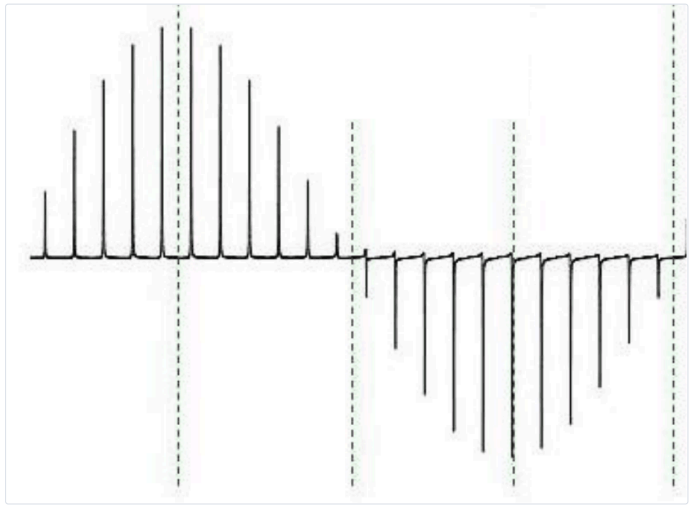
2.12 p1：校准 90° 脉冲 p1: calibrating the 90° pulse

讲义 p.42

p1 是 90° 脉冲的长度，必须校准。实际做法：找到 360° 的零点，再除以 4。
p1 is the length of the 90° pulse and must be calibrated. In practice, find the 360° null and divide by 4.



p1 从小到大排列：信号按正弦变化（用讲义的 p1 = 14.39 µs 计算） 模拟 simulation
Arraying p1: the signal follows a sine (computed with the slide's p1 = 14.39 µs)



真实的 p1 阵列：虚线处分别是过零点 讲义 p.42
A real p1 array: dashed lines mark the zero crossings

实验课讲过要正确设置 90° 脉冲。这个值就是 p1，必须优化，才能得到最大信号（以及最优的相干形成）。

We discussed setting the 90° pulse correctly in the practical. This value is p1 and must be optimized to get max signal (and optimal coherence formation).

做法：**阵列**扫描逐渐增大的 p1，直到找到最大值。

This is done by **arraying** increasing values until we get a maximum.

但通常改为找 **360° 脉冲，再除以 4**。

Usually, the **360° pulse is divided by 4** instead.

算一遍 Worked example

若信号在 57.56 µs 处过零（360°），则 p1 = 57.56 ÷ 4 = 14.39 µs，正好是讲义面板上的值。

If the signal crosses zero at 57.56 µs (360°), then p1 = 57.56 ÷ 4 = 14.39 µs, the value on the slide's panel.

讲义外补充 Beyond the slides

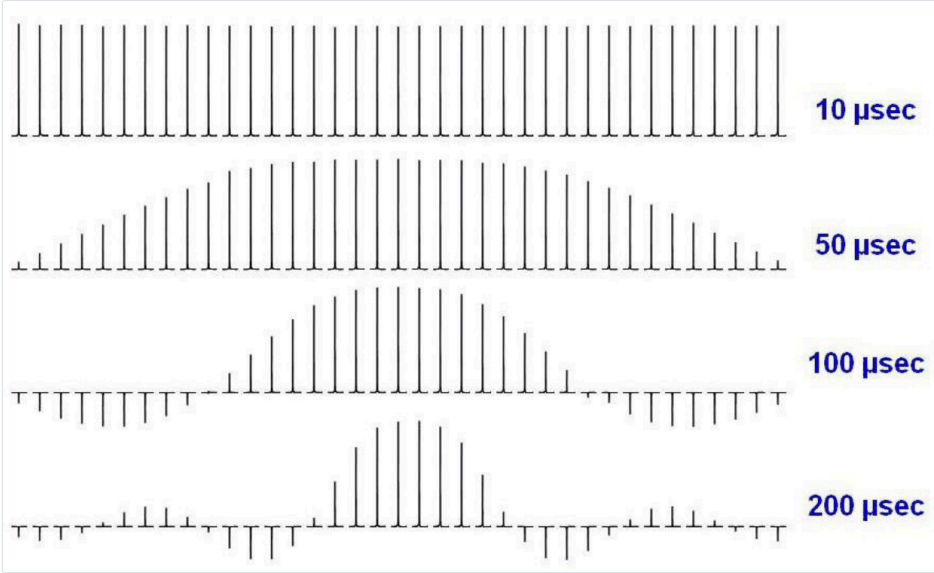
为什么不直接找 90°？最大值处曲线平缓，差几微秒也看不出来；而 360° 处信号陡峭地穿过零点，很好判断。

Why not find 90° directly? The maximum is flat, so a small error goes unseen; at 360° the signal crosses zero steeply and is easy to read.

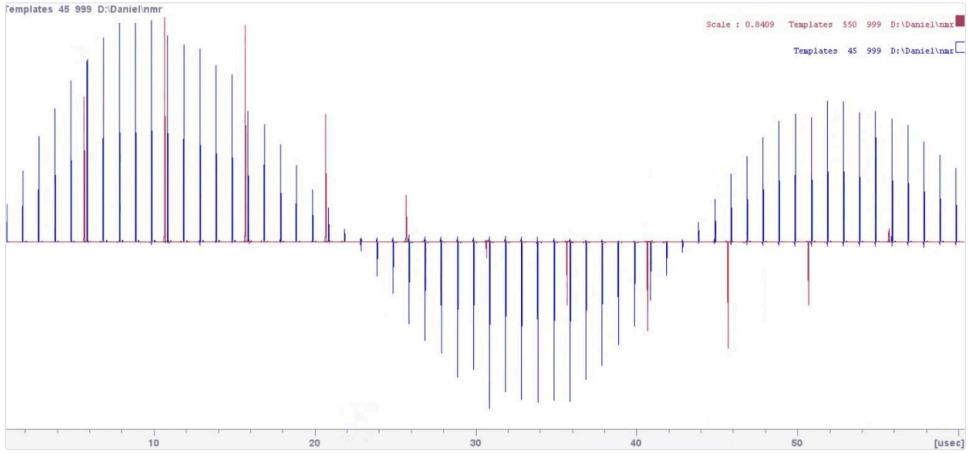
2.13 脉冲长度、带宽和功率 Pulse length, bandwidth and power

讲义 p.43–44

脉冲越短，激发带宽越宽，但需要的功率越高。长度只在特定功率下才正确。
Shorter pulses excite a wider bandwidth but need more power. A length is only correct for a specific power.



脉冲越长，能均匀激发的范围越窄，边上的峰甚至变负 讲义 p.43
Longer pulses: narrower uniform excitation, edge peaks even go negative



不同功率下的 p1 阵列：周期不同，所以“90°”的长度也不同 讲义 p.44
p1 arrays at different powers: the period changes, and so does the “90°” length

p1 也决定**带宽**：能激发到谱图多远的地方。

The value also determines the **bandwidth**: how far away we can excite in the spectrum.

好的 5 mm 探头常见 **约 10–15 µs**。

For a good 5 mm probe, **~10–15 µs** is common.

其它探头可能更长，尤其是大直径探头。

Can be longer for different probes, especially with larger diameters.

长脉冲会降低强度，甚至产生**负强度**。

Long pulses can decrease intensity or even result in **negative intensity**.

另一个参数是**脉冲功率 PLW1**。长度和功率成反比，长度只在特定功率下才正确。

The other parameter is the **pulse power (PLW1)**. Length and power are (inversely) related, and the length is only correct for a specific power.

功率可以改，但（普通 1D）通常不改。它由所用探头决定；短脉冲通常需要更高功率。

It can be changed, but usually is not (for normal 1D). It is determined by the probe; shorter pulses usually need higher power.

讲义外补充 Beyond the slides

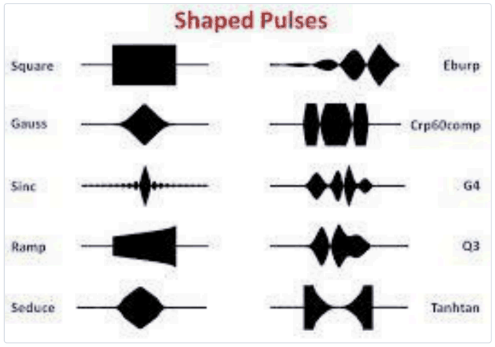
粗略估算：90° 脉冲长度为 p1 时，射频场强度 ≈ 1/(4·p1)。p1 = 14.39 µs 对应约 17 kHz，足以覆盖 10 kHz 的氢谱。

Rough estimate: the RF field strength ≈ 1/(4·p1). p1 = 14.39 µs gives about 17 kHz, enough for a 10 kHz proton spectrum.

3.1 形状脉冲 Shaped pulses

讲义 p.45–46

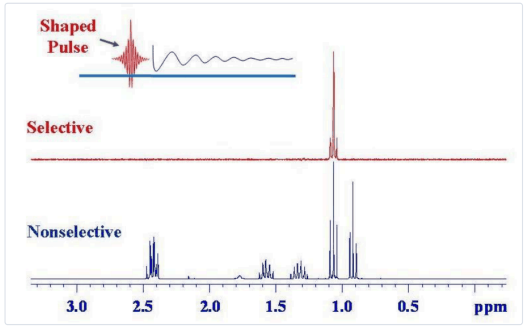
硬脉冲：短、强、覆盖宽。形状（软）脉冲：长、弱、可以只激发想要的峰。
Hard pulses: short, powerful, broad. Shaped (soft) pulses: long, weak, and able to excite only the peaks you want.



常见脉冲形状：Square、Gauss、Sinc、Ramp、Seduce、Eburp、Crp60comp、G4、Q3、Tanhtan

讲义 p.45

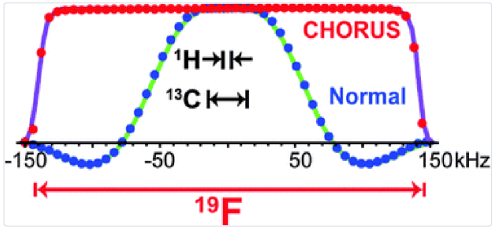
A gallery of pulse shapes



选择性激发只留下一组峰；非选择性激发看到全部

讲义 p.46

Selective excitation keeps one group of peaks; nonselective shows all



普通脉冲（蓝）边缘掉下去；CHORUS（红）在 ±150 kHz 都平坦，盖住整个 ^{19}F 范围

讲义 p.40, 46

A normal pulse falls off at the edges; CHORUS stays flat across ±150 kHz, covering the full ^{19}F range

前面讲的 90° 脉冲叫**硬脉冲**；其它实验可以用**形状（软）脉冲**。

The 90° pulses so far are called **hard pulses**. Other experiments can have **shaped (or soft) pulses**.

工作方式不同：功率要根据想要的长度来设。讲义不细讲，只需知道 p1 以外的脉冲存在。

They work differently: power must be set depending on the desired length. Not covered in detail; just note that pulses other than p1 can exist.

用途：**选择性激发**，比如只关心特定的峰。

Used for **selective excitation**, if you only care about specific peaks.

比硬脉冲长，但所需功率低得多。

Longer than hard pulses, but require considerably less power.

所以探头扛不住高功率时，可能必须用形状脉冲。

So a shaped pulse may be necessary if your probe cannot handle high power.

附带好处：样品加热可能更少。

An ancillary benefit: it may lead to less sample heating.

算一遍 Worked example

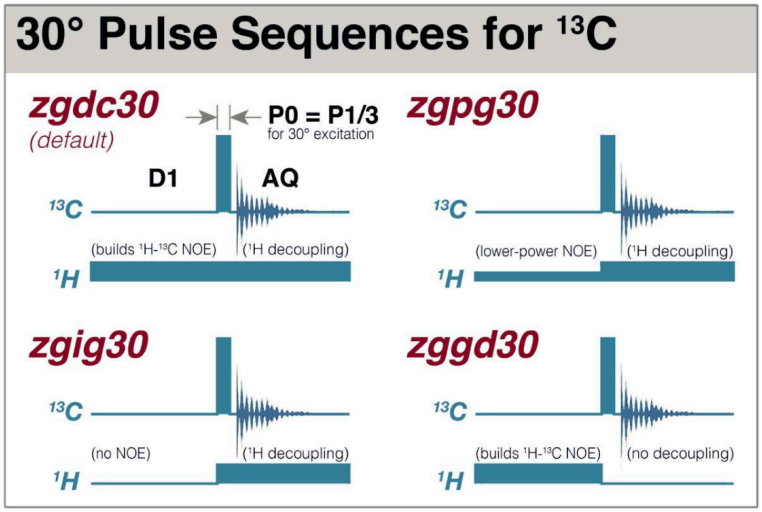
讲义面板里的对比：硬脉冲 P1 = 14.39 µs，10.85 W。形状脉冲 P11 = 80 000 µs，SPNAM1 = Gaus1_270.1000，SPW1 = 9.39 × 10^{-7} W（60.28 dB）。长了约 5 600 倍，功率低了约 1 200 万倍。

From the slides' panels: hard P1 = 14.39 µs at 10.85 W; shaped P11 = 80 000 µs at 9.39 × 10^{-7} W (60.28 dB). About 5 600× longer, about 12 million× less power.

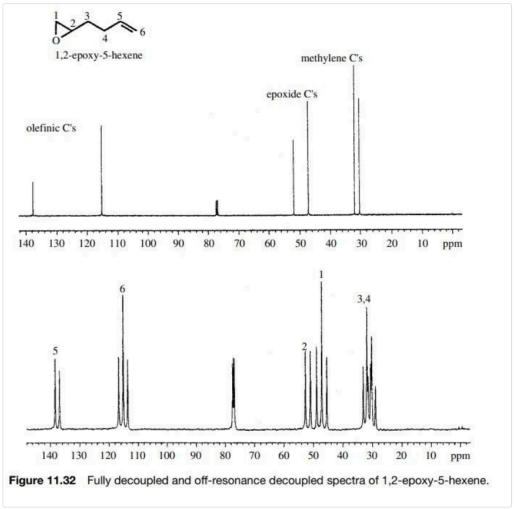
3.2 去耦：四种碳谱序列 Decoupling: four carbon sequences

讲义 p.47–48

AQ 期间去耦让多重峰塌缩；D1 期间照射累积 NOE。定量时不能要 NOE，所以用 zgig。
Decoupling during AQ collapses multiplets; irradiating during D1 builds NOE. Quantitative work cannot have NOE, hence zgig.



四种 30° 碳谱序列：下方色块表示 ¹H 去耦何时打开 讲义 p.48
Four 30° ^{13}C sequences: the blocks show when ^{1}H decoupling is on



上：全去耦，每个碳一条线。下：偏共振去耦，保留部分裂分 讲义 p.47
Top: fully decoupled. Bottom: off-resonance decoupled, keeping some splitting

序列	D1 时 ¹H	AQ 时 ¹H	NOE	用途 Use
zgdc30 默认	开 on	开 on	有 yes	SNR 最高 highest SNR
zgpg30	低功率 low	开 on	有 yes	低功率累积 NOE lower-power NOE
zgig30	关 off	开 on	无 no	**定量必用** required for quantitation
zggd30	开 on	关 off	有 yes	保留耦合 + NOE coupled, with NOE

去耦常用于碳谱和 2D。形状脉冲也可用于去耦。
Decoupling is common in carbon NMR and 2D NMR. Shaped pulses can also be used for decoupling.

去耦的功率和长度都要校准好；功率比发射脉冲低得多。
Power and length of decoupling must be calibrated; the power is much lower than for pulsing.

想要多重峰塌缩，只需在**采集期间**去耦。
To get collapsed multiplets, decoupling is only necessary **during acquisition**.

碳的信号可以通过 **NOE** 增强，所以全程去耦也有用。
For carbon, signal can be enhanced through the **NOE**, so decoupling the whole time can also be useful.

一般 zgdc 的 SNR 最高，但定量工作必须用 zgig。
In general zgdc gives the highest SNR, but zgig must be used for quantitative work.

算一遍 Worked example

讲义面板：CPDPRG2 = waltz65；PCPD2 = 120 μs（去耦序列的 90° 脉冲）；PLW12 = 0.0078519 W（21.05 dB），PLW13 = 0.0039494 W（24.03 dB），都是去耦功率；PLW2 = 0.93444 W（0.29 dB）。比发射脉冲的 10.85 W 低得多。

From the slide's panel: CPDPRG2 = waltz65; PCPD2 = 120 μs (90° pulse of the decoupling sequence); PLW12 and PLW13 are the CPD/BB decoupling powers, 0.0078519 W and 0.0039494 W; PLW2 = 0.93444 W. All far below the 10.85 W pulse.

记忆钩子 Memory hook

名字这样记：dc 全程开 · ig 反门控，只开 AQ · gd 门控，只开 D1 · pg 功率门控。P0 = P1/3 即 30° 激发。

Names: dc on throughout · ig inverse-gated, AQ only · gd gated, D1 only · pg power-gated. P0 = P1/3 gives 30° excitation.

讲义外补充 Beyond the slides

为什么 NOE 破坏定量？不同碳的增强幅度不一样，积分就不再正比于碳数。NOE 在 D1 里慢慢累积，去耦在 AQ 里立刻起效，所以两者可以分开控制。

Why NOE spoils quantitation: enhancement differs between carbons, so integrals no longer track carbon count. NOE builds slowly during D1 while decoupling acts instantly during AQ, so they can be controlled separately.

3.3 额外的延迟和梯度 Additional delays and gradients

讲义 p.49–50

这些值大多已写在脉冲序列里。记住它们的用途，以及梯度重要的是“比值”。
Most of these are built into the pulse sequence. Remember what each is for, and that with gradients what matters is the ratio.

额外的延迟 Additional delays

前面讲的延迟（几乎）每个实验都有；复杂实验还有额外的延迟。

The delays so far are present in (nearly) every experiment; more complex experiments have additional ones.

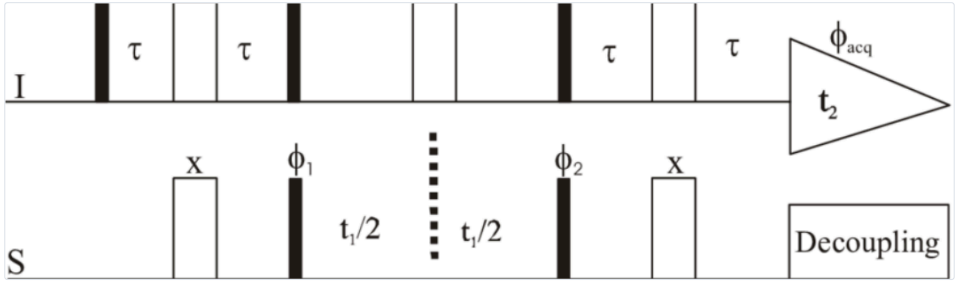
例：W5 压水有**二项式延迟**；HSQC 需要**耦合延迟**。

W5 water suppression has a **binomial delay**; HSQC requires a delay for **coupling**.

有的由用户设，有的根据其它设置自动算出。

Some are set by the user; others are calculated automatically.

参数	讲义里的值	含义 Meaning
D9	30 ms	TOCSY 混合时间 mixing time
D19	0.8 ms	二项式压水延迟 binomial water suppression
CNST2	145 Hz	J(XH) 耦合常数
d4	1.724 ms	等于 1/(4J)，由 CNST2 算出 calculated from CNST2
d0	3 µs	2D 增量延迟 incremented delay
d11	30 ms	磁盘读写延迟 delay for disk I/O
DELTA	6 µs	等于 d0 × 2 + p4，自动计算 calculated automatically



HSQC 序列：τ 延迟让耦合演化，$t_1/2$ 是间接维的增量时间

讲义 p.49

HSQC: τ delays let coupling evolve; $t_1/2$ is the indirect-dimension increment

梯度 Gradients

梯度可以选择**相干转移途径**，或者“清理”谱图。

Gradients can select **coherence transfer pathways** or “clean up” spectra.

梯度之后总要有一个专门的**恢复延迟**，否则谱图会畸变。

Gradients always need a special **recovery delay**, or spectra will be distorted.

梯度也有不同的形状。

Gradients also have different shapes.

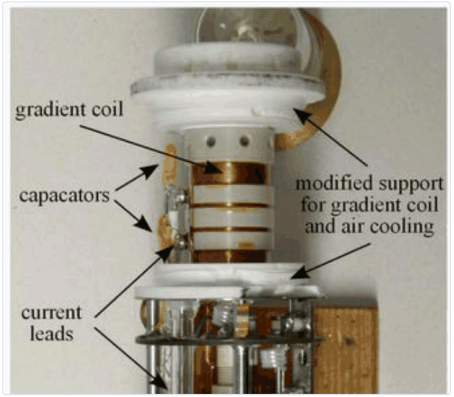
功率和形状通常都已写在脉冲序列里。

Normally both power and shape are given in the pulse sequence.

关键不是绝对值，而是**功率之比**。

What is critical is not the absolute value, but the **ratio of powers**.

参数	讲义里的值	含义 Meaning
D16	0.4 ms	梯度恢复延迟 gradient recovery delay
GPNAM1, 2	SINE.100	梯度形状 gradient shape
GPZ1, 2	34 %, 22 %	梯度强度：看比值 strength: read the ratio
P16	1000 µs	梯度脉冲长度 gradient pulse length



探头里的梯度线圈

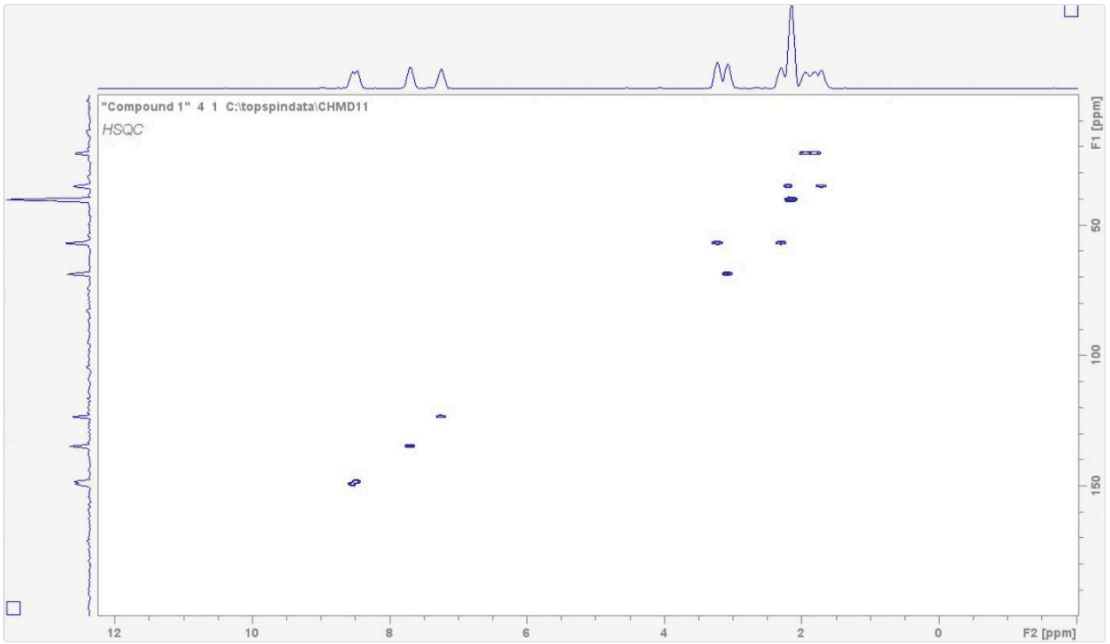
讲义 p.50

The gradient coil inside a probe

4.1 2D：多了一套第二维参数 2D: one more set of parameters

讲义 p.51

2D 的大部分参数和 1D 一样，只是多了一套给第二维；两套都要按实验类型来配。
Most 2D parameters match 1D, with an additional set for the second dimension. Both must be matched to the experiment type.



HSQC：F2 是 ^{1}H（直接维），F1 是 ^{13}C（间接维） 讲义 p.51
HSQC: F2 is ^{1}H (direct), F1 is ^{13}C (indirect)

讲义面板 Panel	F2 (^{1}H)	F1 (^{13}C)
SW 谱宽	14.2774 ppm	199.995 ppm
SWH	7142.857 Hz	25161.491 Hz
IN_F 增量时间	—	39.7375 µs
AQ 采集时间	0.14336 s	0.0050871 s
FIDRES 分辨率	6.975 Hz	196.574 Hz

2D 里大部分参数不变，只是多出一套给第二维。

In 2D most parameters are the same, with an additional set for the second dimension.

这些参数必须与实验类型匹配。

These must be matched to the type of experiment.

例：HSQC 的 SW 约为：氢 **约 20 ppm**，碳 **约 200 ppm**。

For HSQC you may have a SW of **~20 ppm** in proton and **200** in carbon.

算一遍 Worked example

同样的公式在两维都成立：
F2：AQ × 2 × SWH = 2048 点；FIDRES = 7142.9/1024 = 6.98 Hz
F1：AQ × 2 × SWH = 256 点；FIDRES = 25161/128 = 197 Hz
增量时间 IN_F ≈ 1/SWH(F1) = 39.7 µs

The same formulas hold in both dimensions: F2 has 2048 points and F1 has 256. The increment IN_F is about 1/SWH(F1).

讲义外补充 Beyond the slides

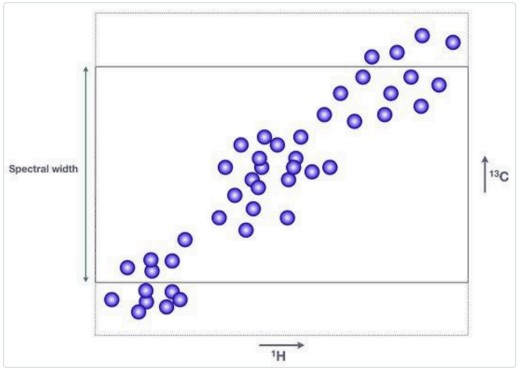
注意 F1 的 FIDRES 比 F2 差了近 30 倍：间接维的每个点都要多跑一整组扫描，付不起那么多点。这就是为什么 2D 总在间接维的分辨率上做文章（4.3，6.3）。

F1′s FIDRES is nearly 30× worse than F2′s: each F1 point costs a whole set of scans, so we cannot afford many. That is why 2D work focuses on F1 resolution (4.3, 6.3).

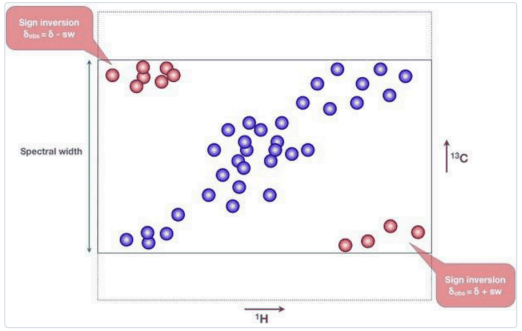
4.2 间接维的 SW 与混叠 SW and aliasing in the indirect dimension

讲义 p.52–53

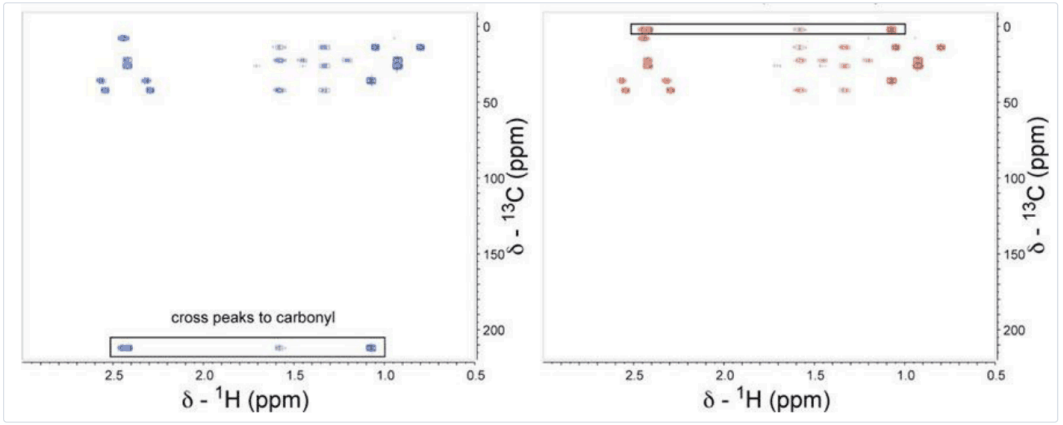
间接维 SW 太小，峰会从窗口一边出去、从另一边回来。这可能是麻烦，也能用来省时间。
If F1′s SW is too small, peaks leave one side of the window and re-enter from the other. This can be a nuisance, or a way to save time.



SW 太小：两端的峰落在窗口之外 p.52
SW too small: peaks at both ends fall outside the window



它们折回窗内（红），符号反转 p.52
They fold back into the window (red), sign-inverted



左：羰基交叉峰在约 200 ppm。右：它们折叠到 0 ppm 附近没有真实峰的区域 讲义 p.53
Left: cross peaks to carbonyl near 200 ppm. Right: they are aliased to near 0 ppm, where no real peaks sit

2D 里 x 轴（**直接维**）照常采集。

y 轴（**间接维**）的采集方式不同。

间接维的 SW 太低，就会看到**混叠**。

坏处：折回来的峰可能和“真实”峰重叠。

好处：能缩短采集时间，做法是让峰折叠到没有真实峰的谱区。

In 2D NMR the x-axis is acquired as normal.

However, the y-axis or **“indirect dimension”** is acquired differently.

In the indirect dimension we can see **aliasing** if SW is too low.

Bad: it can create overlap with “real” peaks.

Useful: it can decrease acquisition time, by aliasing into a spectral region without any real peaks.

算一遍 Worked example

图中标注：折回的峰有**符号反转**，观测位置 $\delta_{obs} = \delta - sw$ 或 $\delta + sw$。所以知道 SW 就能把峰的真实位置算回来。

The figure notes **sign inversion**, with $\delta_{obs} = \delta - sw$ or $\delta + sw$. Knowing SW, you can compute a peak’s true position.

讲义外补充 Beyond the slides

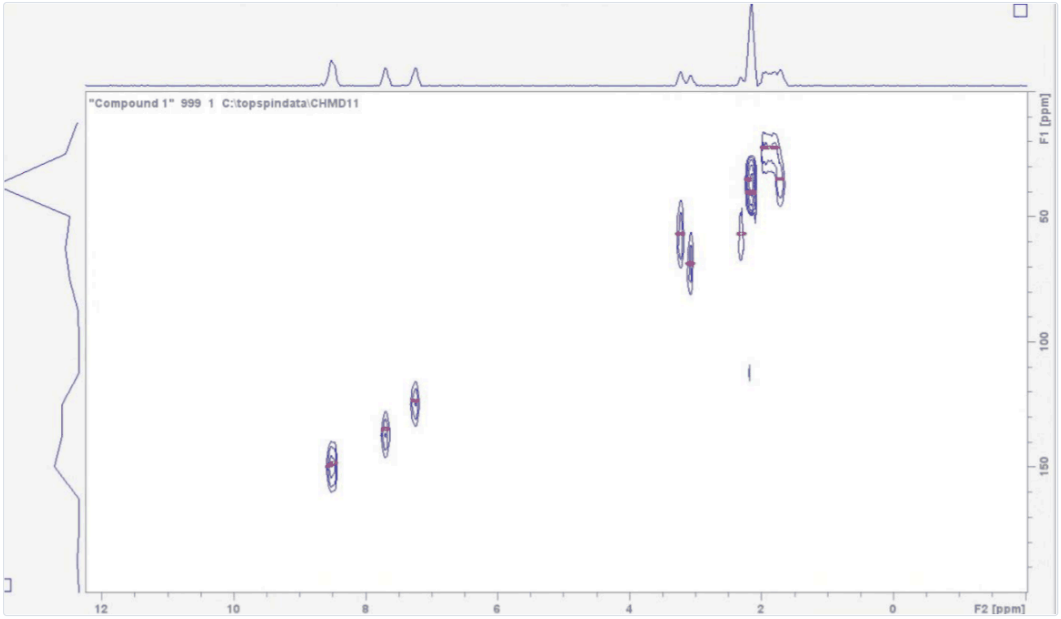
为什么省时间？F1 分辨率 = SWH/(TD/2)。SW 缩小后，同样的分辨率只需要更少的增量。

Why does it save time? F1 resolution = SWH/(TD/2). With a smaller SW, the same resolution needs fewer increments.

4.3 增量数：F1 的 TD Increments: TD in F1

讲义 p.54, 74

F1 的 TD 就是增量数：它决定间接维分辨率，也决定 2D 实验要跑多久。
TD in F1 is the number of increments: it sets the indirect-dimension resolution and how long the 2D experiment takes.



同一数据：256 增量（红）很紧凑；16 增量（蓝）在 F1 方向拉得很长 讲义 p.54
Same data: 256 increments (red) stay compact; 16 increments (blue) smear along F1

TD (F2 | F1)	DS	NS	TD0
512 | 128	16	4	1

2D 还有一个额外的关键参数：**增量数**（F1 的 TD）。
There is an additional key parameter in 2D: the **number of increments** (TD in F1).

它基本上表示每组扫描重复多少次。
It basically indicates how many times each set of scans is repeated.

它影响**间接维的分辨率**。
This value impacts the **resolution in the indirect dimension**.

$$\text{总时间 total time} \approx \text{增量数} \times \text{NS} \times (\text{D1} + \text{AQ} + \ldots)$$

算一遍 Worked example

讲义 p.74：256 增量跑 12.8 分钟，40 增量跑 2.0 分钟。12.8 ÷ 2.0 = 6.4，正好等于 256 ÷ 40。实验时间正比于增量数。

Slide p.74: 256 increments take 12.8 min, 40 take 2.0 min. 12.8 ÷ 2.0 = 6.4 = 256 ÷ 40: time is proportional to increments.

记忆钩子 Memory hook

2D 像拍一部电影：F2 是每一帧的精细程度（很便宜），F1 是总共拍多少帧（每帧都要花整组扫描的时间）。

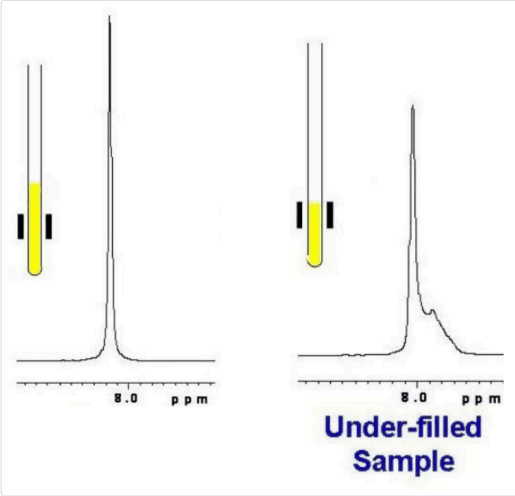
2D is like filming: F2 is the detail in each frame (cheap); F1 is how many frames you shoot (each costs a whole set of scans).

5.1 好谱图从好样品开始 A good spectrum starts with a good sample

讲义 p.55–56

高度要够，不能有固体，要混合均匀。样品有问题，后面什么参数都救不回来。
Fill high enough, no solids, well mixed. No parameter can rescue a bad sample.

上机流程： 样品 Sample → 调谐匹配 Tune & match → 锁场 Lock → 匀场 Shim → 校准脉冲 Calibrate → 设参数 Set parameters



样品量不足：峰形出现明显的拖尾和驼峰 讲义 p.55
Under-filled sample: tailing and a shoulder on the peak

得到好谱图的第一步，是有一个好样品。
The first step to getting a good spectrum is having a good sample.

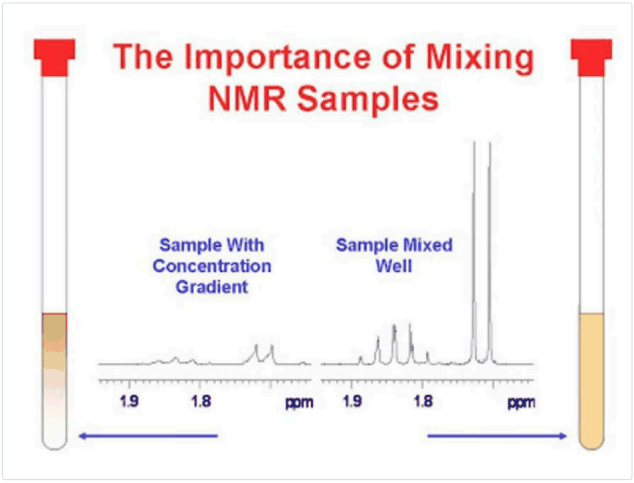
样品要装到合适的高度；只要高于线圈区域，这点相当宽容。
Samples should be filled to an appropriate height, though this is quite forgiving as long as you are above the coil region.

量不够时可以把样品稍微提高一点，但这可能让匀场变差。
If you don't have enough you can try raising the sample a bit, but this can make shims worse.

另一个问题是**沉淀或没混匀**。
Another issue is any sort of **precipitates or poorly mixed samples**.

固体颗粒会改变局部磁化率，导致线形变差。
Solid particles change the local magnetic susceptibility and give poor line shape.

浓度梯度也是问题，常见于粘稠样品或温度变化时。
A **concentration gradient** is another problem, occurring with viscous samples or with temperature changes.

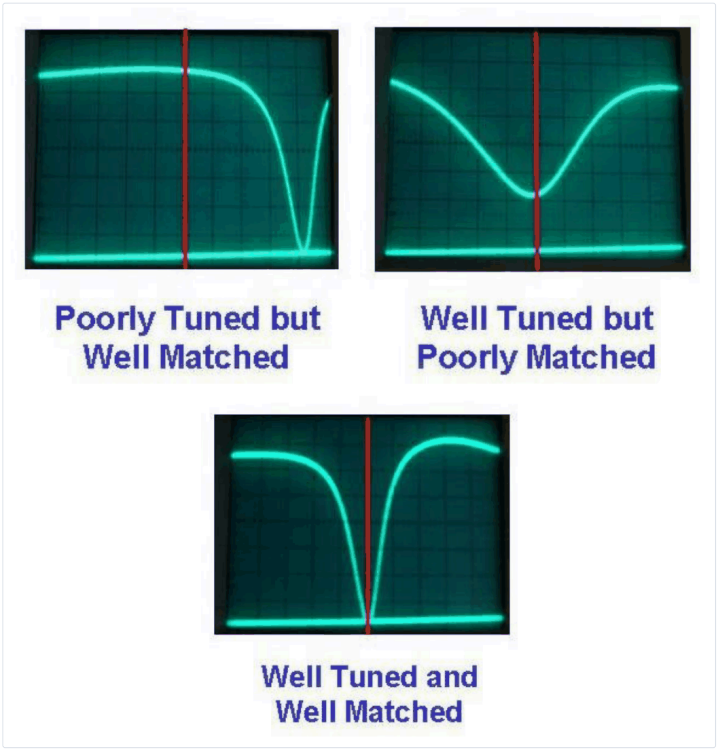


上层和下层浓度不同（左）：谱线一团糟；混匀后（右）：多重峰清晰 讲义 p.56
Concentration gradient (left) vs well mixed (right)

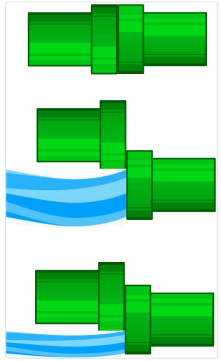
5.2 调谐与匹配 Tuning and matching

讲义 p.57–58

调谐匹配让放大器的功率最大限度地传到样品：调谐是对准频率，匹配是对齐接口。
Tuning and matching transfer maximum power from the amplifier to the sample: tuning finds the frequency, matching lines up the connection.



看曲线：凹陷对准中线 = 调谐好；凹陷够深 = 匹配好 讲义 p.57
Dip on the centre line = tuned; dip reaching the bottom = matched



匹配：管道要对齐 p.58
Matching: line up the pipes



调谐：调收音机频道 p.58
Tuning: set the radio

样品放进磁体后，要做几件准备工作，最关键的是调谐匹配。

Once the sample is in the magnet we need to do a few things to prepare. The most crucial is tuning and matching.

它确保最大功率从放大器传到样品。

This ensures that the maximum power is transferred from the amplifier to the sample.

调谐：基本上就是把收音机调到正确的频率。

Tuning: basically setting your radio to the right frequency.

匹配：稍复杂一点。想象把两段管道对齐：完全对齐时传输最多，往任何一边错开都不好。

Matching: a bit more complicated. Line up the pipes perfectly to get the most transfer; moving too far either way is bad.

曲线 Curve	调谐 Tuned?	匹配 Matched?
凹陷深，但偏离中线	差 poor	好 well
对准中线，但很浅	好 well	差 poor
对准中线，而且很深	好 well	好 well

记忆钩子 Memory hook

调谐看**位置**（左右），匹配看**深度**（上下）。

Tuning is about **position** (left–right); matching is about **depth** (up–down).

讲义外补充 Beyond the slides

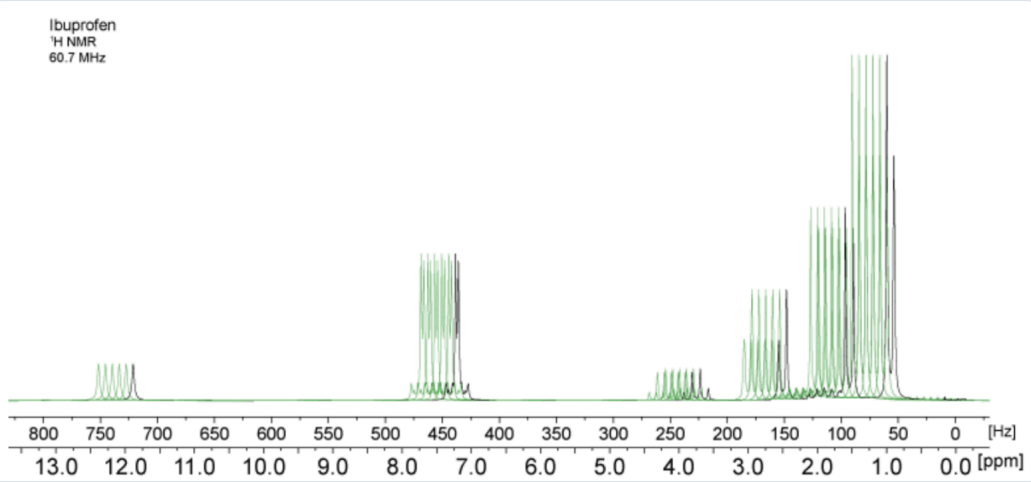
T&M 差的直接后果：发射功率进不去，接收信号也出不来，所以灵敏度下降，同样的 p1 也打不出 90°。

Poor T&M means less power reaches the sample and less signal comes back: lower sensitivity, and the same p1 no longer gives 90°.

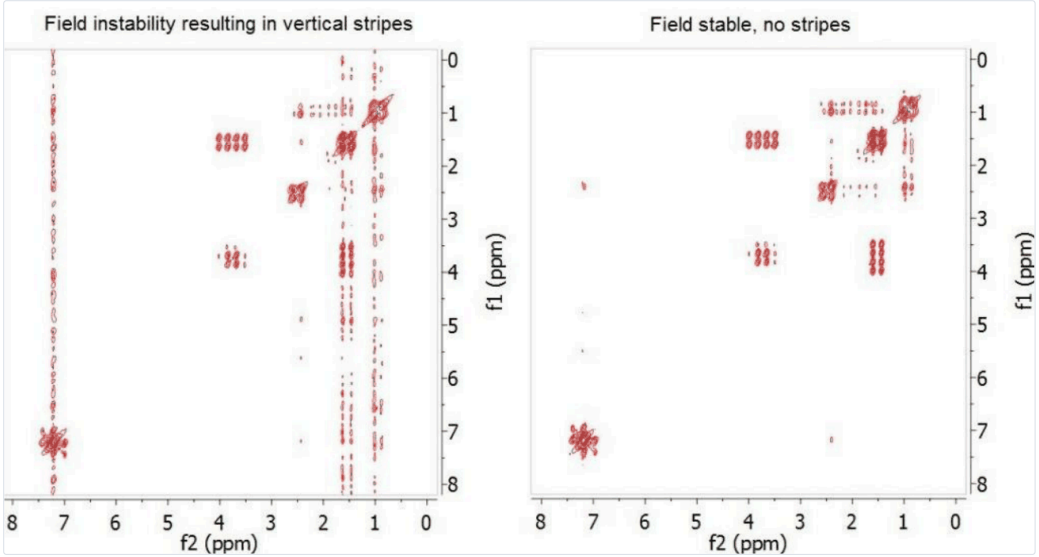
5.3 锁场：为什么需要 Locking: why we need it

讲义 p.59–60

Lock 盯着氘信号，实时补偿磁场随时间的漂移。
The lock watches the deuterium signal and compensates for field drift over time.



布洛芬，60.7 MHz：磁场漂移让多次采集逐渐错位 讲义 p.59
Ibuprofen at 60.7 MHz: field drift pulls repeat acquisitions apart



左：场不稳定产生竖直条纹。右：场稳定，没有条纹 讲义 p.60
Left: field instability gives vertical stripes. Right: stable field, no stripes

NMR 依赖磁场非常**均匀**，而且随时间**稳定**。
NMR relies on the magnetic field being very **homogeneous** and **stable over time**.

任何漂移都会让长时间实验的线形变差。
Any drift leads to poor lineshape over longer experiments.

为防止这一点，要用 **lock**（通常是氘）。
To help prevent this, a **lock** (usually deuterium) is used.

Lock 也有助于减少 2D 伪影：图中的条纹来自场的不稳定，可以由 lock 补偿。
The lock also helps reduce artifacts in 2D NMR: the stripes result from field instability and can be compensated for by a lock.

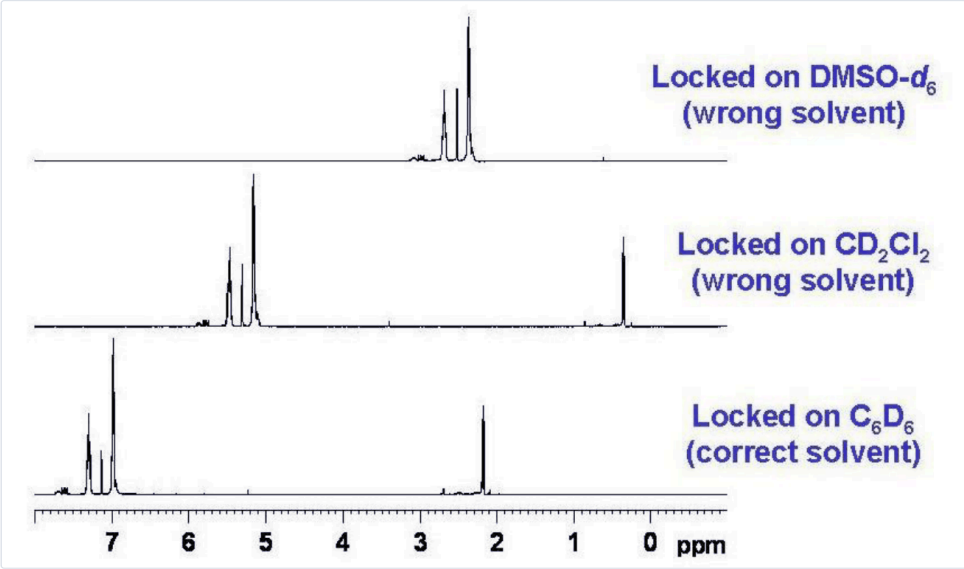
步骤	管什么 Controls	没做好 If poor
T&M	功率 power transfer	信号弱 weak signal
Lock	**时间**上的稳定 stability over time	漂移、条纹 drift, stripes
Shim	**空间**上的均匀 homogeneity in space	宽峰 broad peaks

记忆钩子 Memory hook
T&M 管功率，Lock 管时间，Shim 管空间。这句话能分清三个最容易混的步骤。
T&M is power, lock is time, shim is space. This separates the three steps most often confused.

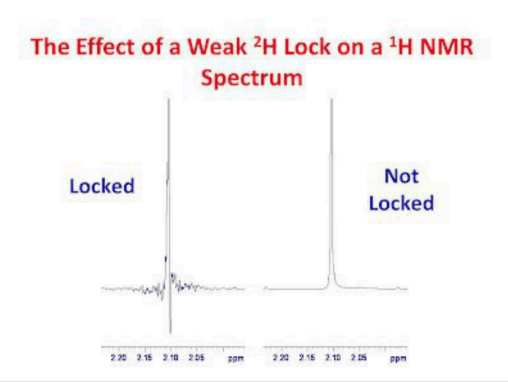
5.4 锁场：选对溶剂，别太弱 Locking: right solvent, strong enough

讲义 p.61–62

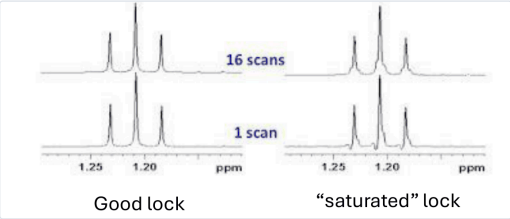
Lock 必须设成你实际用的溶剂；否则可能锁不上，或者化学位移全错。
The lock must be set to the solvent you actually use; otherwise it may not lock, or may give wrong chemical shifts.



C_6D_6 样品：只有设对溶剂时，位移才正确 讲义 p.61
Sample in C_6D_6: only the correct setting gives correct shifts



弱 ^{2}H lock：锁了（左）反而比不锁（右）更畸变 p.62
Weak lock: locked (left) is more distorted than not locked (right)



好的 lock（左）vs “饱和” lock（右） p.62
Good lock (left) vs “saturated” lock (right)

Lock 必须设成你用来锁场的那个化合物。
The lock must be set to the specific compound you are using to lock.

因为谱仪用**溶剂的化学位移**来设定整张谱的化学位移。
The spectrometer uses the **chemical shift of the solvent** to set the chemical shift of your spectrum.

溶剂选错：可能锁不上，也可能给出错误的化学位移。
If chosen incorrectly it may not lock, or might give wrong chemical shifts.

Lock 太弱也会造成谱图畸变；只想用很少量锁场溶剂时就可能发生。
A lock that is too weak can lead to spectral distortions; this can be the case if you only want a small volume of lock solvent.

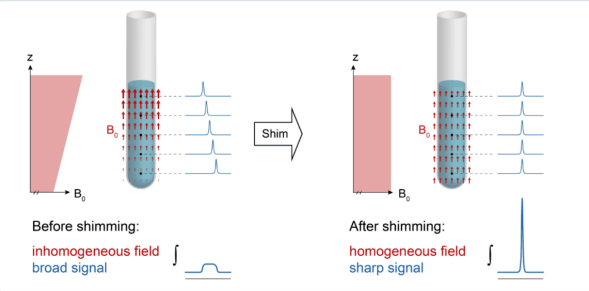
不稳定的 lock 也会影响线形，见右下图的 “饱和” lock。
An unstable lock can also influence lineshape, as with the “saturated” lock at lower right.

出错 What goes wrong
整张谱的位移都偏了，但峰形和间距正常？先查 lock 溶剂设对没有（也可能需要调 SR，见 6.1）。
All shifts offset, but peak shapes and spacing look normal? First check the lock solvent setting (or SR, see 6.1).

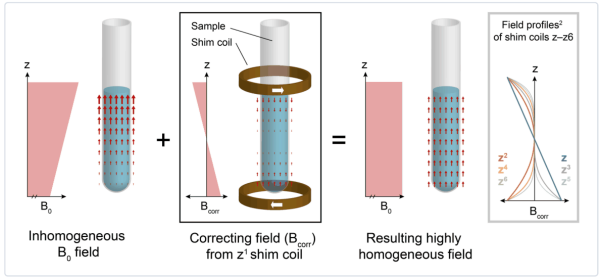
5.5 匀场 Shimming

讲义 p.63–65

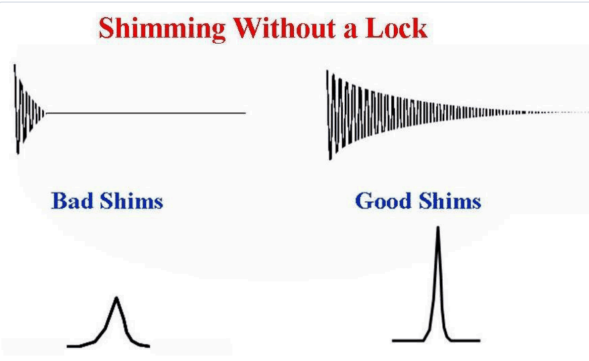
用匀场线圈微调磁场，抵消每个样品不同的不均匀性，让谱线变尖。
Shim coils slightly adjust the field to cancel each sample's inhomogeneity and give sharp lines.



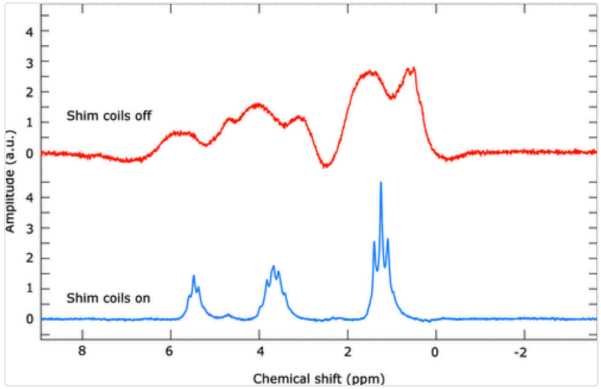
不均匀 → 各处频率不同 → 宽峰。匀场后 → 尖峰 p.63
Inhomogeneous: broad. After shimming: sharp



不均匀场 + z^1 线圈的校正场 = 均匀场 p.63
Inhomogeneous B_0 + correcting field = homogeneous



匀场差：FID 很快衰减，峰宽。匀场好：FID 长，峰尖 p.64
Bad shims: short FID, broad peak. Good: long FID, sharp peak



关掉匀场（红）：裂分全没了。打开（蓝）：裂分清楚 p.65
Shims off (red): splitting gone. On (blue): clear

希望样品所在区域非常均匀。但每个样品都不同，所以要匀场来调整这些差异，得到尖锐的线形。

We want the sample region very homogeneous. Each sample differs, so shimming adjusts for these differences and gives sharp lineshape.

做法：用**匀场线圈**稍微改变磁场。

This is done by slightly changing the magnetic field using **shim coils**.

匀场非常重要，通常是自动的**梯度匀场**。

Shimming is super important and is usually automated **gradient shimming**.

有些探头没有梯度，只能**手动匀场**：通常看曲线下面积增大，它对应更多信号。

Some probes lack gradients and must shim **manually**, usually by looking for increased area under the curve, which corresponds to more signal.

匀场不好，会丢失关键的裂分（也就是耦合）信息。

Without good shimming you can lose key splitting (and thus coupling) information.

记忆钩子 Memory hook
合唱团：大家音高一致，声音才集中（尖峰）；各唱各的，声音就散开（宽峰）。

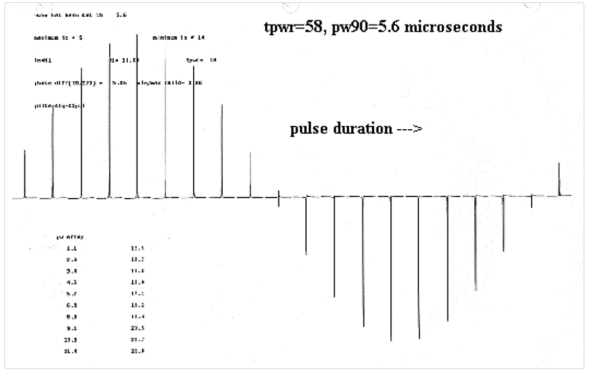
A choir: everyone on the same pitch gives a focused sound (sharp peak); scattered pitches smear out (broad peak).

5.6 校准脉冲，然后设参数 Calibrate, then set parameters

讲义 p.66–68

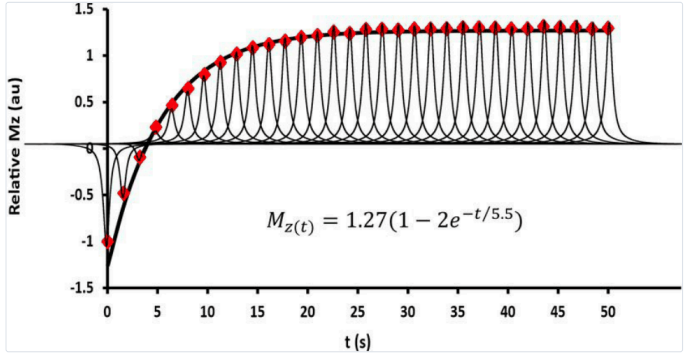
先校准 p1，其它形状脉冲都由它算出来，p1 错则全错。然后再设 O1P、NS、D1。

Calibrate p1 first: shaped pulses are calculated from it, so if p1 is wrong, everything is wrong. Then set O1P, NS and D1.



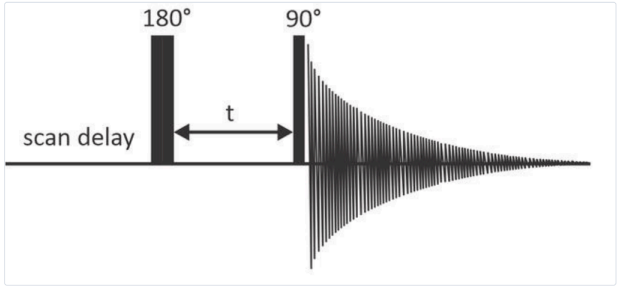
真实的 90° 校准阵列（此例 pw90 = 5.6 µs） 讲义 p.66

A real 90° calibration array (here pw90 = 5.6 µs)



反转恢复数据拟合：$M_z(t) = 1.27(1 - 2e^{-t/5.5})$，所以 T1 = 5.5 s 讲义 p.68

Inversion recovery fit: $M_z(t) = 1.27(1 - 2e^{-t/5.5})$, so T1 = 5.5 s



测 T1 的序列：180° — t — 90°，改变 t 讲义 p.68

The T1 sequence: 180° — t — 90°, varying t

准备好采集后，校准所需的脉冲。从 90° 脉冲开始，其它形状脉冲都据此重新计算。	Once ready to acquire, calibrate any required pulses. Start with the 90° pulse; all other shaped pulses are recalculated from it.
碳脉冲通常不是每个样品都校准（尤其是“反式”探头）。	Carbon pulses are usually not calibrated on every sample (especially for “inverse” probes).
形状脉冲不用实验校准，而是根据 p1 计算。**p1 必须正确，否则其它全错。**讲义 p.67 的探头参数表：^{1}H 11.00 µs, 0.93444 W；^{13}C 17.00 µs，5.6522 W。	Shaped pulses are not calibrated experimentally but calculated from p1. **It is critical to have correct p1 values or everything else will be wrong.** Probe table on p.67: ^{1}H 11.00 µs, 0.93444 W; ^{13}C 17.00 µs, 5.6522 W.
然后设参数，这会因样品和实验类型差别很大。	Then the desired parameters are set; this varies greatly with sample and experiment.
O1P 很关键，尤其是压溶剂时。	**O1P** is critical, especially for solvent suppression.
NS 取决于需要多少 SNR、样品多浓。	**NS** depends on how much SNR you need and how concentrated your sample is.
可能还要先测样品的 **T1**。	This may also involve finding the **T1** of your sample.

算一遍 Worked example

T1 = 5.5 s，1D 的 D1 = 3–5 × T1 = 16.5–27.5 s。信号过零点在 t = T1 · ln 2 ≈ 3.8 s，与左图一致。

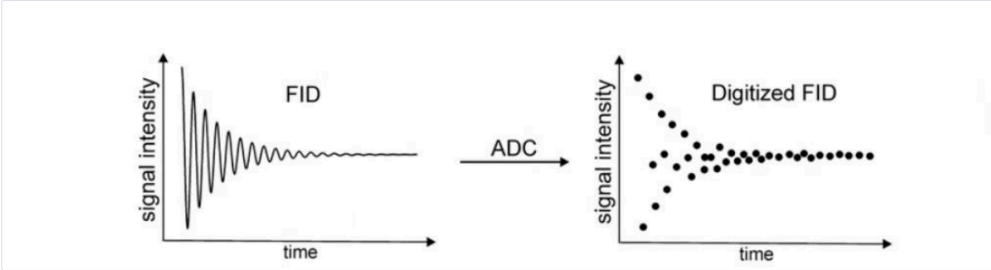
With T1 = 5.5 s, a 1D D1 is 16.5–27.5 s. The signal crosses zero at t = T1 · ln 2 ≈ 3.8 s, matching the plot.

6.1 从 FID 到谱图 From FID to spectrum

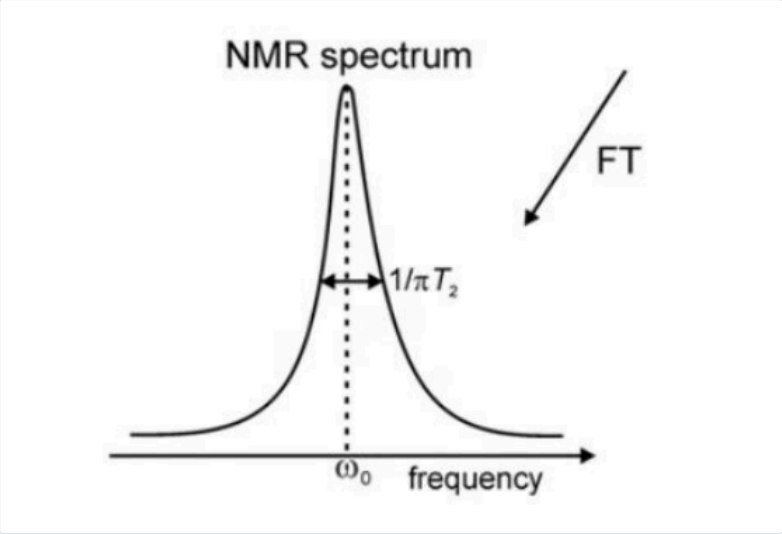
讲义 p.69–71

FID 本身没什么用，我们要的是频率谱图。处理参数同样要按样品调整。
The FID isn't that useful; we want a frequency spectrum. Processing parameters must also be adapted to the sample.

实际计算顺序： FID → 线性预测 LP → 窗函数 Window → 补零 ZF → FT → 相位 Phase → 基线 Baseline → 定标 Reference （讲义的讲解顺序略有不同）



FID 经 ADC 变成一串点 讲义 p.69
The FID becomes a series of points through the ADC



FT 得到谱图；线宽 = 1/(πT$_2$) 讲义 p.69
FT gives the spectrum; linewidth = 1/(πT$_2$)

前面讲的都是从信号到 FID。但 FID 不太有用，我们想要的是频率谱图。

What we discussed so far goes from signal to FID. But the FID isn't really that useful: we want a frequency spectrum.

有一整套参数必须按样品类型调整。

An entire set of parameters must be adapted to the type of sample.

数据采完后可用别的软件处理；这对小众应用或自动化工作流更有用。

Acquired data can be processed in different software: more useful for niche applications or workflow automation.

两大公司：Mestrelab（**Mnova**）和 **ACD/Labs**。讲师认为 UTSC 有 Mnova 校园授权。开源的有 **NMRPipe**。

Two big companies: Mestrelab (**Mnova**) and **ACD/Labs**. The lecturer believes UTSC has a campus licence for Mnova. Open source: **NMRPipe**.

第一个关键参数是 **SI**：FT 用的点数，通常设为 TD 的倍数。

The first key parameter is **SI**: the number of points used for the FT, normally a multiple of TD.

SR：调整参考频率（例如根据内标）。这里另一个有用的参数就是它。

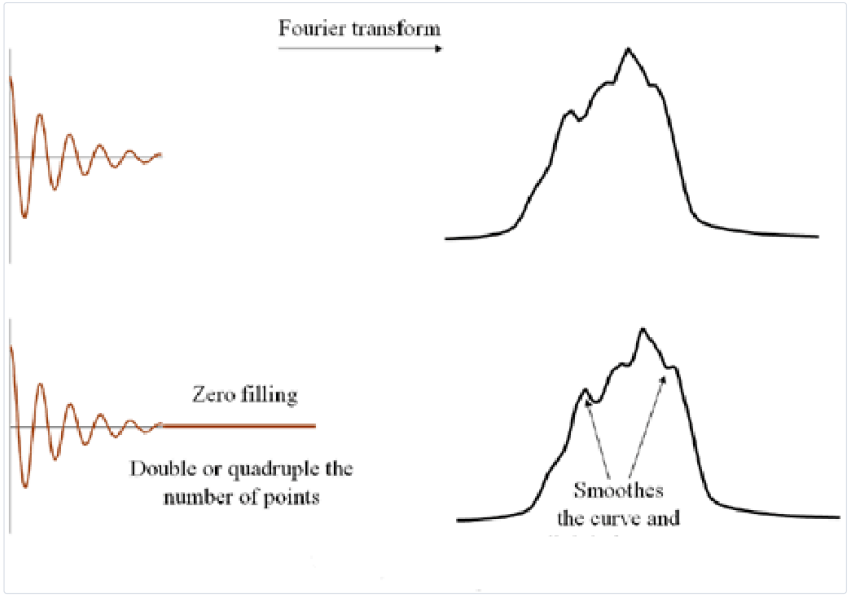
SR is the only other useful one here: it adjusts the reference frequency if needed (e.g., based on an internal standard).

参数	讲义里的值	含义 Meaning
SI	16384	FT 用的点数 size of real spectrum
SF	80.141 MHz	谱仪频率 spectrometer frequency
OFFSET	14.99891 ppm	低场边界 low-field limit
SR	0 Hz	参考频率调整 reference frequency
HZpPT	0.098444 Hz	每点 Hz 数 = SWH/SI

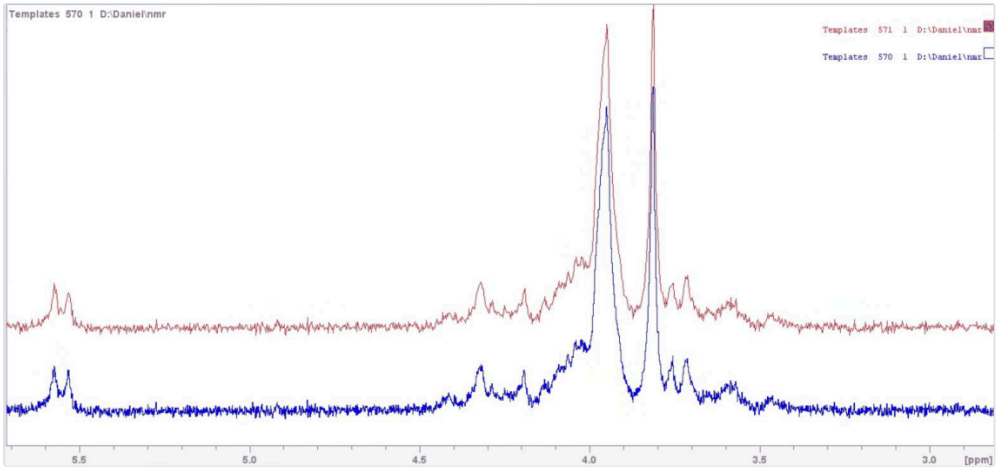
6.2 补零 Zero filling

讲义 p.72

补零就是在 FID 尾部加 0：谱线更平滑，但没有增加任何新信息。
Zero filling appends zeros to the FID: smoother lines, but no new information.



补零让曲线更平滑，把点数翻倍或变四倍 讲义 p.72
Doubling or quadrupling the points smooths the curve



真实谱图：有没有补零，差别通常很小 讲义 p.72
Real spectra: usually very little difference

容易混淆的一点：设 **SI = TD**，实际用的点数是 TD 的**两倍**。

Confusingly, setting **SI = TD** actually uses **double** the number of points compared to TD.

多出来的点作为 0 接在末尾。

The additional points are appended to the end as zeros.

这叫**补零**（zero filling），它能提高“数字分辨率”。

This is called **zero filling** and improves the “digital resolution”.

实际上差别通常很小。

In practice there is usually very little difference.

算一遍 Worked example

讲义的 80 MHz 例子：TD = 16384，SWH = 1612.9 Hz。FID 分辨率 = 1612.9/8192 = 0.197 Hz；SI = 16384 时每点 0.098 Hz。点距正好缩小一半。

The slides’ 80 MHz example: TD = 16384, SWH = 1612.9 Hz. FID resolution is 0.197 Hz; with SI = 16384, 0.098 Hz per point: exactly half.

讲义外补充 Beyond the slides

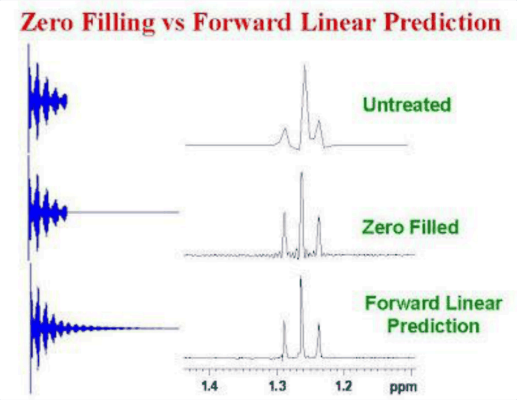
为什么 SI = TD 等于翻倍？TopSpin 的 TD 同时数实部和虚部的点，SI 只数实部。所以 SI = TD 时，实部点数已是采集所得的两倍。

Why does SI = TD double the points? In TopSpin, TD counts real and imaginary points together; SI counts real points only.

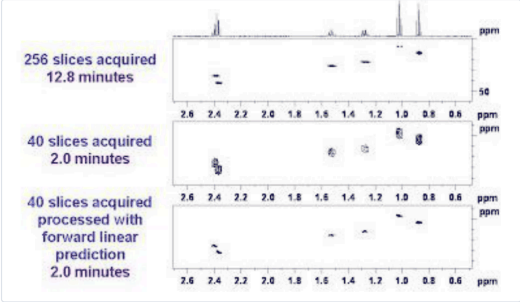
6.3 前向线性预测与 2D Forward linear prediction, and in 2D

讲义 p.73–74

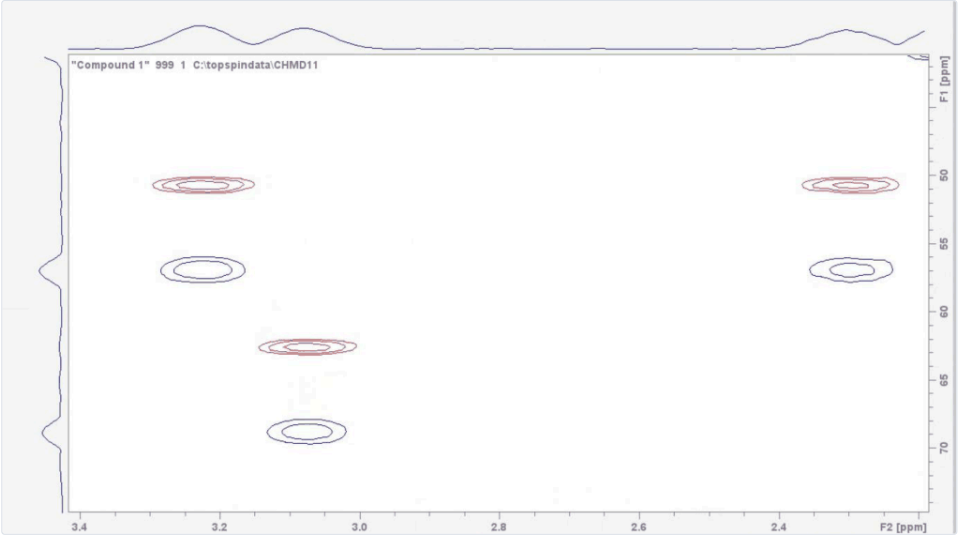
线性预测根据已有的 FID 外推出更多点，比补零更能救截断；但这些点是“假”的。
Linear prediction extrapolates more points from the existing FID, rescuing truncation better than zero filling; but those points are “fake”.



截断的 FID：不处理、补零、前向 LP p.73
Untreated, zero filled, forward LP



256 增量 12.8 分钟；40 增量 2.0 分钟；40 增量 + LP 也是 2.0 分钟 p.74
256 slices 12.8 min; 40 slices 2.0 min; 40 + LP still 2.0 min



增量较多时，LP 的效果更细微（红 vs 蓝） 讲义 p.74
With more increments the effect is more subtle (red vs blue)

补零不增加任何信息，主要只是让谱图好看。	Zero filling does not add any additional information; it mainly just makes the spectrum look nicer.
FID 被截断时，补零能帮一点，但线性预测可能更好。	If the FID is truncated, zero filling can help a bit, but linear prediction may be a better option.
线性预测：根据当前 FID 外推，人为延长 FID。	It artificially extends the FID based on an extrapolation of the current FID.
线性预测在 **2D** 中更常见，因为能省很多时间。	Linear prediction is typically more common in **2D**, as it can save lots of time.
增量很少时效果最大，可以接近长得多的实验的分辨率；增量多时效果较细微。	The effect is largest with few increments and can approach the resolution of much longer experiments; it is more subtle with more increments.

出错 What goes wrong

这是在加“假”数据，可能带来伪影。出现的新峰要先怀疑是不是 LP 造出来的。	This adds “fake” data and can lead to artifacts. Be suspicious of any new feature that LP might have created.

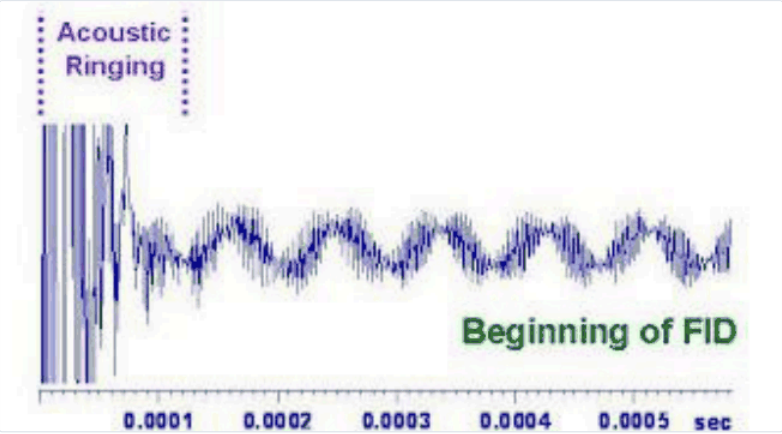
记忆钩子 Memory hook

补零是把照片放大（没有新细节）；LP 是让 AI 根据现有画面“补全”（可能很像，也可能编造）。	Zero filling is enlarging a photo (no new detail); LP is extending it from what’s already there (often right, sometimes invented).

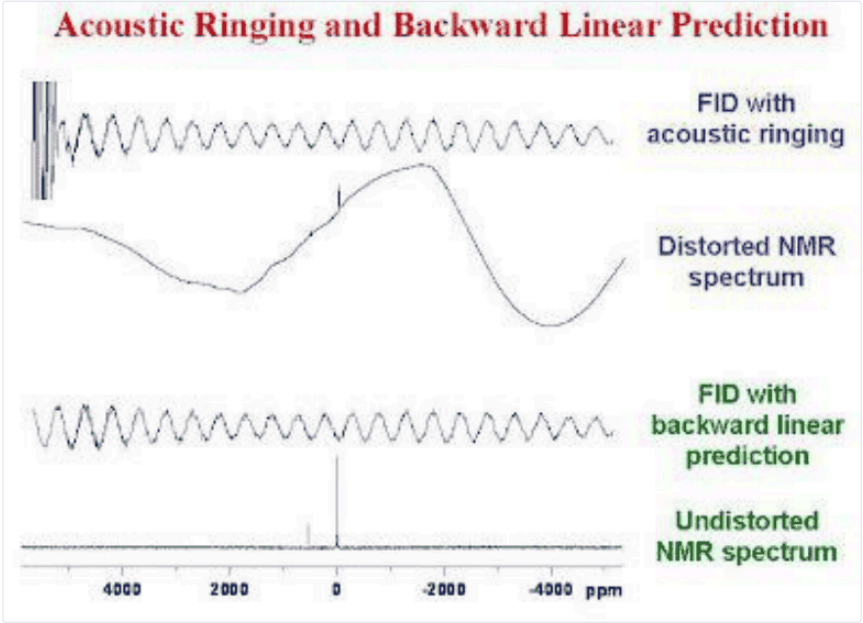
6.4 后向线性预测 Backward linear prediction

讲义 p.75

向后预测到采集开始之前，修复 FID 开头的问题，从而救回基线。
Predicting backwards, to before acquisition started, repairs the start of the FID and rescues the baseline.



FID 的开头：前几个点被声学振铃污染 讲义 p.75
The beginning of the FID, contaminated by acoustic ringing



有振铃：FID 开头异常，谱图基线大起伏。后向 LP 之后：基线变平 讲义 p.75
With ringing: distorted spectrum. After backward LP: undistorted

线性预测也可以向后做，预测到采集开始之前。

It is also possible to linearly predict backwards to before acquisition started.

这有助于解决**声学振铃**（acoustic ringing）这类造成基线差的问题。

This can help resolve issues such as **“acoustic” ringing** that lead to poor baselines.

也有助于调整**群延迟**（group delay），但这点没那么重要。

It also helps adjust for the **group delay**, but that is less critical.

讲义外补充 Beyond the slides

为什么 FID 开头错一点，基线就坏？FID 开头的点对应谱图里最“慢”的整体起伏。这几个点错了，就给整张谱叠加了一条弯曲的基线。

Why do a few bad points ruin the baseline? The first FID points govern the slow, broad features of the spectrum. Corrupt them and a rolling baseline is added everywhere.

讲义外补充 Beyond the slides

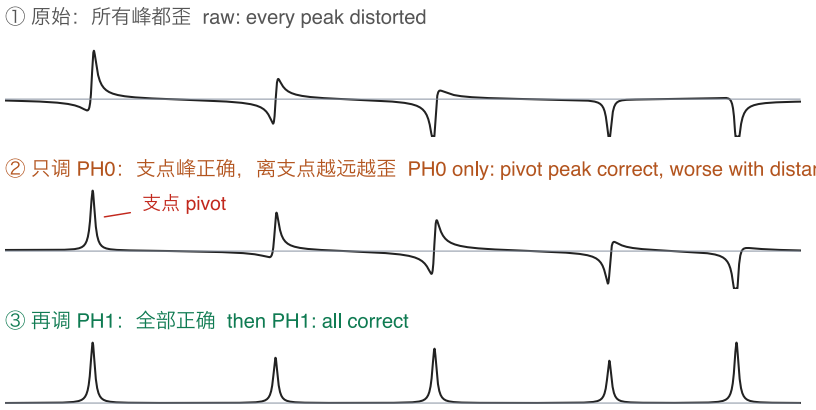
声学振铃：强射频脉冲让探头线圈发生机械振动，振动再感应出假信号。在低频核（低旋磁比）上更明显。

Acoustic ringing: the strong RF pulse makes the probe coil vibrate mechanically, and the vibration induces a false signal. It is worse for low-frequency nuclei.

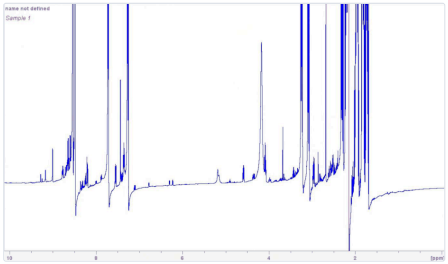
6.5 相位校正 Phase correction

讲义 p.76–77

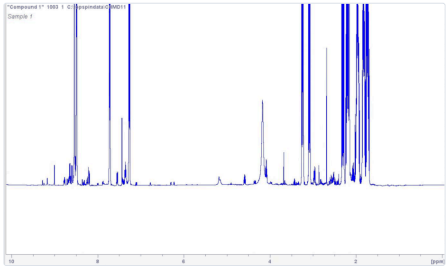
PH0 给所有峰加同一个角度；PH1 按峰到支点的距离加不同的角度。
PH0 adds the same angle to every peak; PH1 adds a different angle depending on the distance from the pivot.



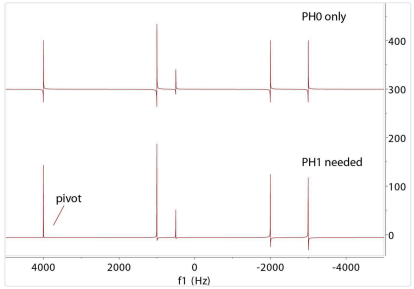
相位校正的两步 模拟 simulation
The two steps of phase correction



相位错误：基线歪、有负强度 p.76
Improperly phased: negative intensity



相位正确 p.76
Properly phased



上：所有峰歪得一样，PH0 就够。下：支点的峰已正，越远越歪，需要 PH1 讲义 p.77
Top: every peak equally distorted, PH0 is enough. Bottom: the pivot peak is correct, others worsen with distance, so PH1 is needed

不正好在 O1P 的峰，相位会略有不同。
Peaks that are not exactly at O1P will have a slightly different phase.

必须校正，才能得到好谱图。
This must be corrected in order to get a good spectrum.

相位不对就会有**负强度**，很难处理。
If improperly phased, there will be **negative intensity**, which makes it difficult to work with.

零阶 PH0：与化学位移无关，给每个峰加同样的量。
Zero order: independent of the chemical shift; adds the same amount to each peak.

一阶 PH1：给每个峰加不同的量，取决于**支点**放在哪里。
First order: adds a different amount to each peak based on where you put the **pivot point**.

记忆钩子 Memory hook
PH0 是把所有人一起转；PH1 是拧麻花：支点不动，离支点越远，转得越多。
PH0 turns everything together. PH1 is a twist: the pivot stays still and the further away, the more it turns.

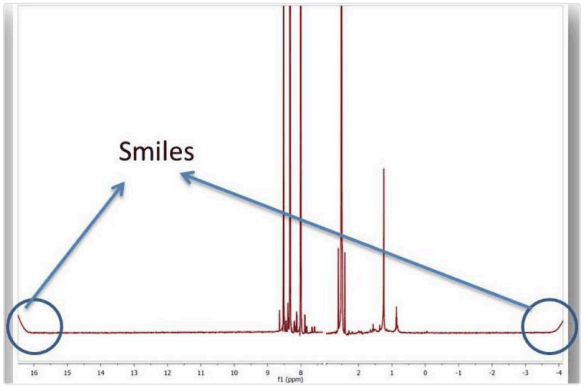
讲义外补充 Beyond the slides
手动操作：把支点放在一个大峰上，用 PH0 调正它；再看另一端的峰，只用 PH1 调正。
By hand: put the pivot on a big peak and fix it with PH0; then go to a far peak and fix it with PH1 only.

6.6 基线问题：先认症状 Baseline problems: know the symptoms

讲义 p.78–80

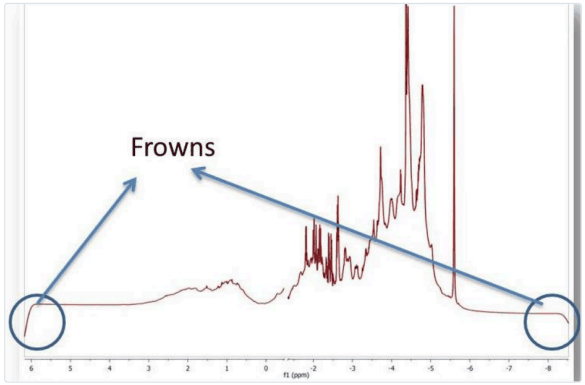
基线不平会把峰带到负区、让积分不准。常见三类：边缘弯曲、整体弯曲、宽探头背景。

An uneven baseline pushes peaks negative and makes integrals inaccurate. Three common kinds: curved edges, overall curve, broad probe background.



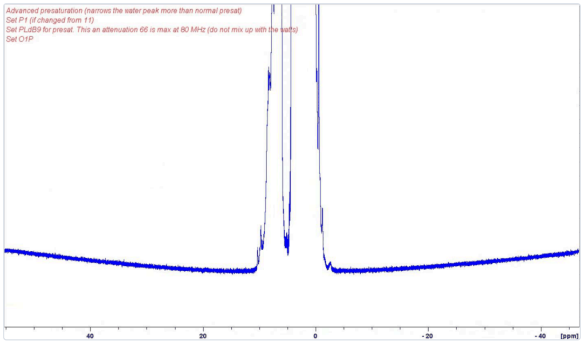
Smile：两端上翘 p.79

Smile: edges curve up



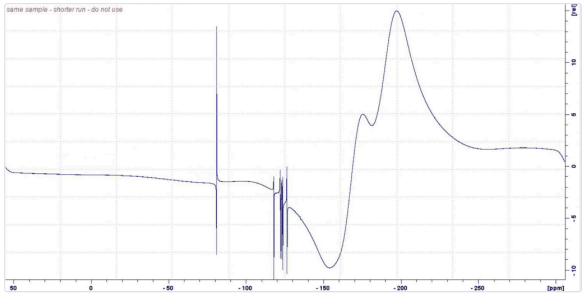
Frown：两端下弯 p.79

Frown: edges curve down



整体弯曲：多项式很好修 p.80

Overall curve: easy with a polynomial



宽探头背景：难得多 p.80

Broad probe background: much harder

NMR 谱的基线常常不完全平。

这会让有些峰跑到负区，降低**积分准确度**；可用基线校正修复。

Smile / frown：谱图边缘向上或向下弯。单独出现时很好修，线性校正就够。

整体弯曲：多项式函数很容易修。

宽探头背景：不只是基线问题。可以校正，但难得多。

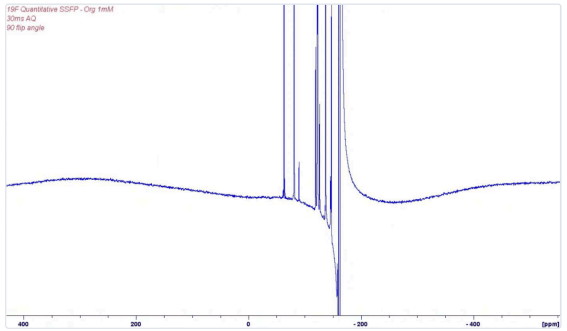
Often the baseline will not be completely flat.

This can push peaks into the negative region, decreasing the **accuracy of integrals**. Baseline correction fixes it.

Smiles or frowns: the edge of the spectrum curved up or down. Very easy to fix, even with just a linear correction.

A curve in the baseline: easily fixed by a polynomial function.

Broad probe background: not just a baseline issue. It can be corrected but is much more challenging.



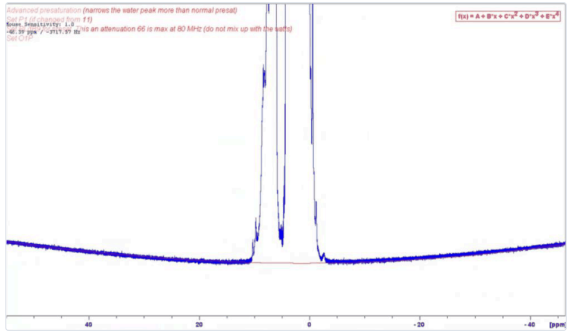
一个非常差的基线：峰被带到负区 讲义 p.78

A very bad baseline: peaks pushed into the negative region

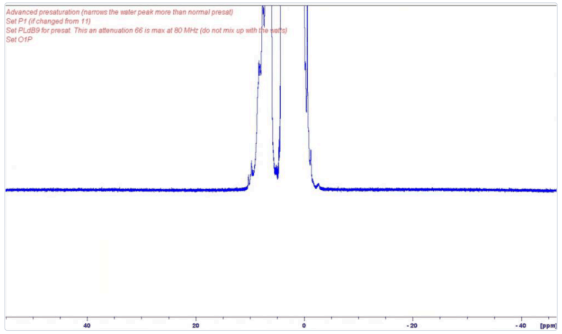
6.7 基线校正：用什么工具 Baseline correction: the tools

讲义 p.78, 81–82

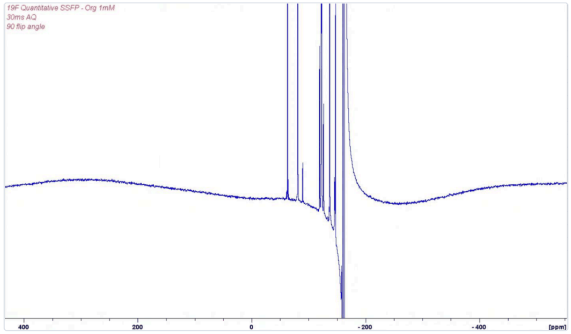
最常用多项式或正弦函数；想省事有自动算法；难的谱交给样条。
Polynomials and sinusoids are most common; automated algorithms save effort; hard spectra call for splines.



校正前：红线是五次多项式拟合的基线 讲义 p.81
Before: a fifth-degree polynomial fit of the baseline



校正后：基线变平 讲义 p.81
After: flat baseline



难例：^{19}F 谱。讲义留作现场演示 讲义 p.82
A tricky ^{19}F spectrum, demonstrated live

基线校正种类很多，最常见的是**多项式或正弦函数。**

例：对谱图用五次多项式，得到校正后的谱。

自动或半自动方法：**Whittaker Smoother、非对称重加权惩罚最小二乘。**

讲师的经验：Mnova 的基线校正更好，尤其是样条（splines）；但多数谱在 TopSpin 里就足够了。

There are many varieties, but the most common are **polynomials or sinusoidal functions**.

Example: a fifth-degree polynomial applied to the spectrum, and the corrected result.

Automated or semi-automated: the **Whittaker Smoother**, **asymmetrically reweighted penalized least squares**.

The lecturer finds Mnova does better with baselines, especially using splines; however, most are more than fine in TopSpin.

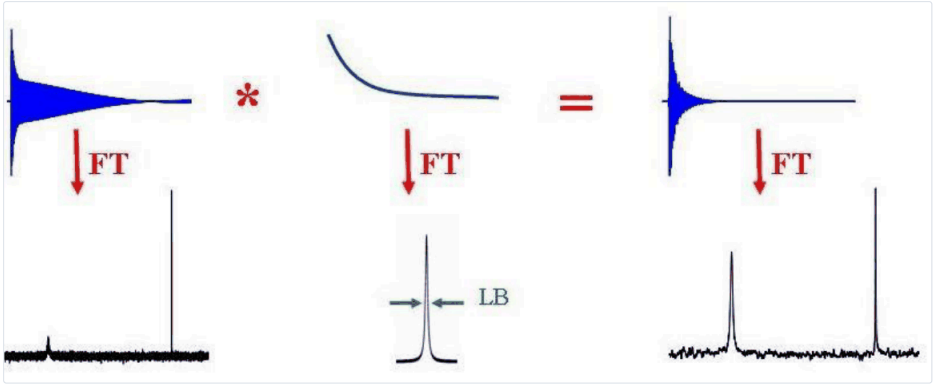
参数	讲义里的值	含义 Meaning
ABSG	5	abs 的多项式次数（0–5） polynomial degree
ABSF1	6.05451 ppm	abs 的左边界 left limit
ABSF2	2.54078 ppm	abs 的右边界 right limit
BCFW	0.15 ppm	bc 的滤波宽度 filter width
COROFFS	0 Hz	校正偏移 correction offset
BC_mod	no	FID 基线模式 FID baseline mode

6.8 窗函数与线宽展宽 Window functions and line broadening

讲义 p.83–84

FT 前给 FID 乘一个函数。1D 常用指数函数：压低噪声多的尾部，SNR 升高，但谱线变宽。

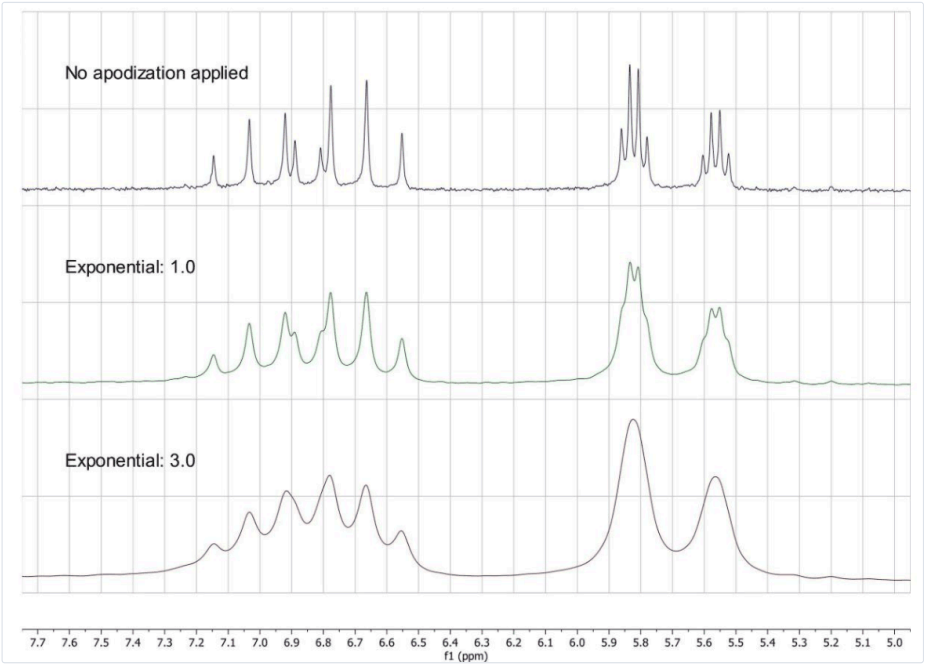
Multiply the FID by a function before FT. In 1D an exponential is common: it suppresses the noisy tail, raising SNR but broadening lines.



FID × 指数函数：在频率域相当于每条线都加宽 LB

讲义 p.83

FID × exponential: in frequency, every line broadens by LB



LB = 0、1.0、3.0：噪声更少，但裂分逐渐糊掉

讲义 p.84

LB = 0, 1.0, 3.0: less noise, but splitting blurs away

直接做 FT 通常不是最好的结果。

Usually just doing the FT does not give optimal results.

要对 FID 施加**切趾 / 窗函数**，得到更好的谱图。

Instead, **apodization or a window function** is applied to the FID to get a better spectrum.

1D 里最常用**指数函数**，配一个相应的**线宽展宽 LB**。

In 1D it is most common to use an **exponential function** with a corresponding **line broadening (LB)**.

LB 应设得**小于你最尖峰的线宽**。

This value should be set **lower than the sharpest peak you have**.

分辨率和灵敏度之间要小心平衡。

There is a delicate balance between resolution and sensitivity.

有机样品氢谱通常 **0–0.5 Hz**；宽峰样品可以更高；碳谱大得多（**5–15 Hz**）。

Usually **0–0.5 Hz** is good for an organic sample (proton). Higher for broad samples. Carbon is usually much larger (**5–15 Hz**).

参数	讲义里的值	含义 Meaning
WDW	QSINE, QSINE	窗函数类型 window function
LB	0.30 Hz	em 的线宽展宽 line broadening
GB	0, 0.1	gm 的高斯极大位置 Gaussian max position
SSB	2	正弦钟移位 sine bell shift
TM1, TM2	0.1, 0.9	梯形窗的左右界 trapezoid limits

记忆钩子 Memory hook

FID 前段信号多，后段噪声多但带着细节。强调前段 → 灵敏度；强调后段 → 分辨率。

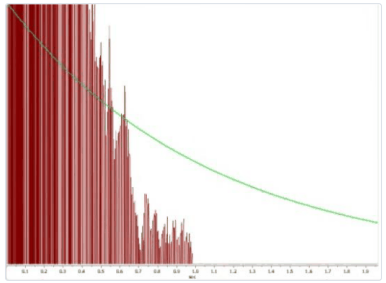
The early FID is mostly signal; the late FID is noisier but carries fine detail. Weight early for sensitivity, late for resolution.

6.9 窗函数一览：HSQC 和 COSY 怎么选 Window gallery: HSQC vs COSY

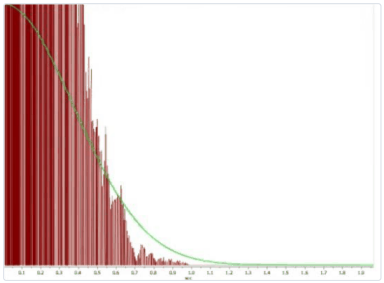
讲义 p.85–86

选哪个窗，取决于谱图以及要 SNR 还是分辨率。必记：HSQC 用移位 sine²，COSY 用不移位 sine²。

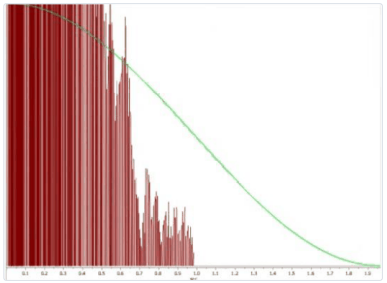
The right window depends on the spectrum and on whether you want SNR or resolution. Must know: HSQC, shifted sine²; COSY, unshifted sine².



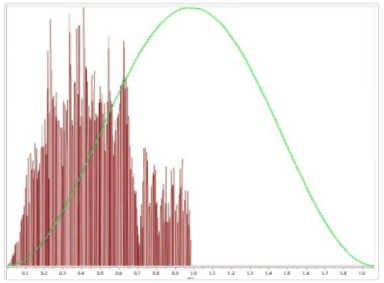
指数 Exponential p.85



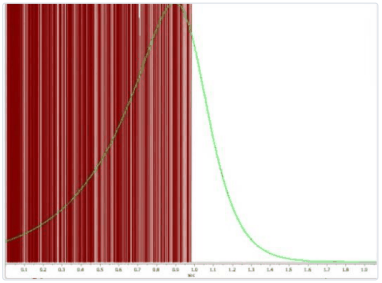
高斯 Gaussian p.85



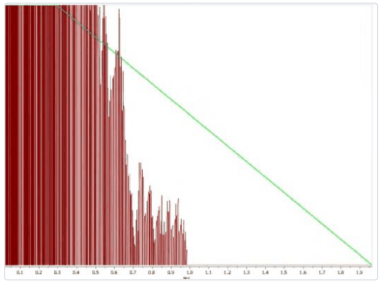
移位 sine² Shifted p.85



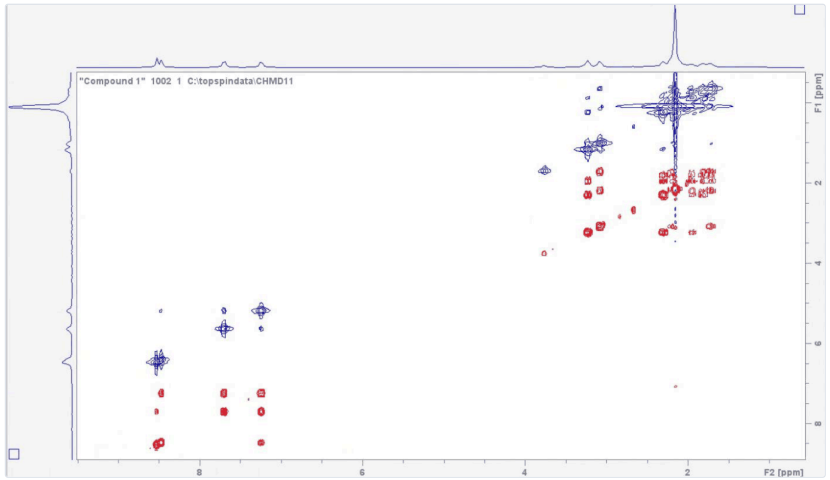
不移位 sine² Unshifted p.85



Traficante p.85



梯形 Trapezoidal p.85



COSY：正确处理（红，不移位）vs 用了移位 sine（蓝，峰拖出十字形长尾） 讲义 p.86

COSY processed correctly (red, unshifted) vs with a shifted sine (blue, with long cross-shaped tails)

正确的窗函数取决于谱图，以及要 SNR 还是分辨率。

The correct one depends on the spectrum and whether you want SNR or resolution.

HSQC：两维通常都用**移位 sine²**。

HSQC: a **sine squared shifted** function is usually appropriate in both dimensions.

COSY：用**不移位 sine²**。

COSY: an **unshifted sine squared** function is used.

记忆钩子 Memory hook

看曲线起点：移位 sine² 从最大值开始，保留信号，给 HSQC。不移位 sine² 从 0 开始，砍掉 FID 开头，给 COSY。

Look at the start: shifted sine² begins at its maximum, keeping signal, for HSQC. Unshifted sine² begins at 0, killing the early FID, for COSY.

讲义外补充 Beyond the slides

TopSpin 里：WDW = QSINE 即 sine²，SSB = 2 是移位，SSB = 0 是不移位。讲义 p.83 的面板正是 QSINE + SSB 2。

In TopSpin: WDW = QSINE means sine²; SSB = 2 is shifted, SSB = 0 unshifted. The panel on p.83 shows QSINE with SSB 2.

7.1 看到什么 → 哪里错了 → 怎么办 Symptom → cause → fix

讲义 全讲义

看到什么 What you see	原因 Likely cause	怎么办 Fix	节 §
强峰两侧有波纹 Wiggles either side of strong peaks	FID 被截断：TD / AQ 太小 Truncated FID: TD or AQ too small	提高 TD；已采完可用 LP Raise TD; if already acquired, use LP	2.2
峰宽、裂分没了，但没有波纹 Broad peaks, no splitting, no wiggles	TD 太小，或匀场差，或 LB 太大 Low TD, bad shims or too much LB	看 FID：是被切断，还是衰减太快? Check the FID: cut off, or decaying too fast?	2.2, 5.5
积分不准 Integrals are wrong	D1 太短 / 截断 / 基线差 / 碳谱有 NOE Short D1, truncation, baseline, carbon NOE	D1 ⩾ 5 × T1；碳谱用 zgig；校正基线 D1 ≥ 5 × T1; zgig for carbon; fix baseline	2.8, 3.2
单扫就畸变，基线怪，有假峰 Distortion on one scan, odd baseline, fake peaks	RG 太高，ADC 溢出 RG too high, ADC overflow	下调 RG，或用 rga Lower RG, or use rga	2.5
谱形正常，但小峰看不见 Clean spectrum, small peaks missing	RG 太低或 NS 太少 RG too low or NS too few	提高 RG；增加 NS（4 倍 → SNR 2 倍） Raise RG; raise NS (4× gives 2× SNR)	2.6, 2.9
预期的峰不见了 Expected peaks missing	SW 没盖住，或 O1P 太偏 SW too narrow or O1P off	调 O1P、加宽 SW，必要时分段扫描 Move O1P, widen SW, scan in segments	2.4, 2.11
离中心远的峰变弱或变负 Far peaks weak or negative	脉冲带宽不够 Pulse bandwidth too narrow	缩短 p1；O1P 居中；宽带脉冲 Shorter p1; centre O1P; broadband pulse	2.13
水峰没压掉 Water not suppressed	O1P 没设在溶剂上 O1P not on the solvent	把 O1P 设到溶剂共振 Set O1P to the solvent resonance	2.11
F1 维有符号相反、位置奇怪的峰 Odd sign-inverted peaks in F1	间接维混叠 Aliasing in the indirect dimension	加宽 F1 的 SW，或按 δ ± sw 算回 Widen F1 SW, or back-calculate δ ± sw	4.2
2D 的峰沿 F1 拉得很长 2D peaks smeared along F1	增量太少 Too few increments	增加 F1 的 TD，或用 LP Raise TD in F1, or use LP	4.3, 6.3

7.1 看到什么 → 哪里错了 → 怎么办（续） Symptom → cause → fix (continued)

讲义 全讲义

看到什么 What you see	原因 Likely cause	怎么办 Fix	节 §
线形拖尾、有驼峰、很宽 Tailing, shoulders, broad lines	样品不足、有颗粒、浓度梯度、匀场差 Under-filled, solids, gradient, bad shims	补足液面、过滤或混匀、重新匀场 Top up, filter or mix, re-shim	5.1, 5.5
整张谱位移都偏了 All shifts offset	Lock 溶剂选错，或参考没校准 Wrong lock solvent, or reference off	设对溶剂；调整 SR Set the right solvent; adjust SR	5.4, 6.1
2D 有竖直条纹 Vertical stripes in 2D	场不稳定 Field instability	用 lock 补偿 Compensate with a lock	5.3
峰脚扭曲，线形怪 Distorted peak bases	Lock 太弱或不稳定 Weak or unstable lock	增加锁场溶剂，稳定 lock More lock solvent, stabilise the lock	5.4
信号普遍偏弱 Everything is weak	调谐匹配差，或 p1 不对 Poor T&M or wrong p1	重新调谐匹配，重新校准 p1 Redo T&M and calibrate p1	5.2, 2.12
基线大幅慢起伏 Large rolling baseline	声学振铃 Acoustic ringing	后向 LP Backward LP	6.4
谱图边缘上翘或下弯 Spectrum edges curve up or down	Smile / frown Smile or frown	线性基线校正 Linear baseline correction	6.6
一个宽大的驼峰叠在基线上 A broad hump under the spectrum	探头背景 Probe background	高阶或样条校正（较难） High-order or spline correction (hard)	6.6, 6.7
峰一半朝上一半朝下，有负强度 Half-up, half-down peaks; negative intensity	相位错误 Phase error	先 PH0，再 PH1 PH0 first, then PH1	6.5
处理后噪声太大 / 峰太糊 Too noisy / too blurred after processing	LB 太小 / 太大 LB too small / too large	调 LB：^{1}H 0–0.5 Hz，^{13}C 5–15 Hz Tune LB: ^{1}H 0–0.5 Hz, ^{13}C 5–15 Hz	6.8
COSY 的峰拖出十字形长尾 COSY peaks with long cross-shaped tails	窗函数选错 Wrong window function	用不移位 sine^{2} Use unshifted sine^{2}	6.9

7.2 参数词典 Parameter dictionary

参数	中文	English
PULPROG	脉冲程序	Current pulse program
TD	FID 点数	Size of FID
SW, SWH	谱宽 ppm, Hz	Sweep width
AQ	采集时间	Acquisition time
RG	接收增益	Receiver gain
DW	驻留时间	Dwell time between points
DE	预扫描延迟	Pre-scan delay
D1	弛豫（循环）延迟	Relaxation (recycle) delay
DS	空扫数	Dummy scans, data discarded
NS	扫描数	Number of scans
TD0	外层循环数	Multiplies the number of scans
SFO1	实际工作频率	Frequency of channel 1
BF1	基频	Base frequency
O1, O1P	偏置 Hz, ppm	Offset; spectrum centre
NUC1	观测核	Nucleus for channel 1
P1	90° 硬脉冲长度	High-power 90° pulse
PLW1	脉冲功率	Power level for pulse
P11	90° 形状脉冲长度	90° shaped pulse
SPNAM1	形状脉冲文件	File name for shaped pulse
SPOAL1	频偏相位对齐	Phase alignment of freq. offset
SPOFFS1	形状脉冲频偏	Offset frequency for shaped pulse
SPW1	形状脉冲功率	Shaped pulse power
CPDPRG2	去耦程序	Decoupling program, e.g. waltz65
PCPD2	去耦 90° 脉冲	90° pulse for decoupling sequence
PLW12, 13	去耦功率	Power for CPD/BB decoupling

参数	中文	English
D9	TOCSY 混合时间	TOCSY mixing time
D19	二项式压水延迟	Binomial water suppression delay
CNST2	耦合常数 J(XH)	J(XH) for coupling delays
d0	2D 增量延迟	Incremented delay in 2D
d4	1/(4J) 延迟	1/(4J) delay
d11	磁盘读写延迟	Delay for disk I/O
D16	梯度恢复延迟	Gradient recovery delay
GPNAM, GPZ	梯度形状、强度	Gradient shape, strength
P16	梯度脉冲长度	Homospoil/gradient pulse
IN_F	间接维增量时间	Increment for delay in F1
FIDRES	FID 分辨率	FID resolution
SI	FT 点数	Size of real spectrum
SR	参考频率调整	Spectrum reference frequency
OFFSET	低场边界	Low-field limit of spectrum
HZpPT	每点 Hz 数	Spectral resolution
SPECTYP	谱图类型	Type, e.g. COSY, HMQC
WDW	窗函数	Window function
LB	线宽展宽	Line broadening for em
GB	高斯极大位置	Gaussian max position for gm
SSB	正弦钟移位	Sine bell shift
TM1, TM2	梯形窗边界	Limits for tm
ABSG	多项式次数	Polynomial degree for abs
ABSF1, 2	abs 左右边界	Left and right limits for abs
BCFW, COROFFS	滤波宽度、校正偏移	Filter width; correction offset
BC_mod	FID 基线模式	FID baseline mode

7.3 术语表与讲义勘误 Glossary and slide corrections

acoustic ringing	声学振铃	dynamic range	动态范围	matching	匹配
acquisition	采集	FID	自由感应衰减	mixing time	混合时间
aliasing	混叠 / 折叠	Fourier transform	傅里叶变换	Nyquist frequency	奈奎斯特频率
analogue signal	模拟信号	gradient	梯度	overflow	溢出
apodization	切趾	group delay	群延迟	phase correction	相位校正
baseline correction	基线校正	hard pulse	硬脉冲	pivot point	支点
bandwidth	带宽	heteronucleus	杂核	polynomial	多项式
binomial delay	二项式延迟	homogeneity	均匀性	pre-amplifier	前置放大器
chemical shift	化学位移	increment	增量	precipitate	沉淀
coherence	相干	indirect dimension	间接维	quantitative	定量
coherence transfer pathway	相干转移途径	inverse gated	反门控	receiver gain	接收增益
concentration gradient	浓度梯度	inverse probe	反式探头	recycle delay	循环延迟
cross peak	交叉峰	line broadening	线宽展宽	shaped pulse	形状脉冲
decoupling	去耦	lineshape	线形	shimming	匀场
digital resolution	数字分辨率	linear prediction	线性预测	solvent suppression	溶剂压制
dummy scan	空扫	lock	锁场	truncation	截断
dwell time	驻留时间	magnetic susceptibility	磁化率	tuning	调谐

讲义里要留意的 4 处小错 Four slips on the slides

p.32：7 × T1 写作约 99.991%。1 − e^{-7} = 99.91%，所以应是 99.9%。

p.32: 7 × T1 is given as ~99.991%. 1 − e^{-7} = 99.91%, so it should read 99.9%.

p.23：“continuous digital signal” 应为 continuous analogue signal。

p.23: “continuous digital signal” should be continuous analogue signal.

p.6：Nyquist 那句漏了一个词，应为 “highest frequency that can be **represented** given a sample rate”。

p.6: a word is missing; it should be “highest frequency that can be **represented** given a sample rate”.

p.70：“Mestrenova” 是软件名，公司是 Mestrelab；“advanced chemical development labs” 通常写作 ACD/Labs。

p.70: Mestrenova is the software, Mestrelab the company; “advanced chemical development labs” is ACD/Labs.

自测题 Self-test questions

先遮住答案，自己说出来。想不起来的那题，才是你真正要复习的地方。
Say each answer out loud before you check it. The ones you cannot recall are exactly what to review.

1	为什么前置放大器要靠近探头?	Why is the pre-amp placed close to the probe?	§ 1.1
2	16 位 ADC 有多少级? 8 位呢?	How many levels does a 16-bit ADC have? 8-bit?	§ 1.2
3	采样率 20 kHz，能正确表示的最高频率是多少? 更高的频率会怎样?	At a 20 kHz sample rate, what is the highest frequency represented? What happens above it?	§ 1.3
4	TD = 32768，SWH = 8000 Hz。求 DW 和 AQ。	TD = 32768, SWH = 8000 Hz. Find DW and AQ.	§ 2.1
5	TD 太小有哪几种表现?	What are the symptoms of too low a TD?	§ 2.2
6	TD 不变，SW 加宽，AQ 和分辨率怎么变?	If SW is widened at fixed TD, what happens to AQ and resolution?	§ 2.1
7	RG 太低和太高，哪个让谱图失真? 另一个有什么后果?	Which distorts the spectrum, RG too low or too high? What does the other cost?	§ 2.5–2.6
8	怎么设 RG? 碳谱呢?	How do you set RG? For carbon?	§ 2.6
9	T1 = 2 s，D1 = 5 × T1。D1 是多少? 恢复多少磁化?	T1 = 2 s, D1 = 5 × T1. What is D1, and how much magnetization recovers?	§ 2.7
10	为什么不总用很长的 D1?	Why not always use a long D1?	§ 2.8
11	想把 SNR 翻倍，NS 要变成几倍?	To double SNR, by what factor must NS rise?	§ 2.9
12	SFO1、BF1、O1 的关系? O1 和 O1P 有什么区别?	How do SFO1, BF1 and O1 relate? How do O1 and O1P differ?	§ 2.10
13	O1P 设错有哪三个后果?	Name three consequences of a wrong O1P.	§ 2.11
14	为什么常用 360° 除以 4 来定 p1?	Why is p1 often found as the 360° pulse divided by 4?	§ 2.12
15	长脉冲有什么问题? p1 和功率什么关系?	What is the problem with long pulses? How do p1 and power relate?	§ 2.13
16	形状脉冲和硬脉冲相比有什么特点? 何时必须用?	How do shaped pulses compare with hard pulses? When might you need one?	§ 3.1
17	碳谱定量用哪个序列? 它的去耦时段是什么?	Which carbon sequence is for quantitation, and when does it decouple?	§ 3.2
18	梯度最关键的是什么? 之后必须加什么?	What matters most with gradients, and what must follow them?	§ 3.3
19	J = 145 Hz，d4 是多少?	With J = 145 Hz, what is d4?	§ 3.3
20	HSQC 的典型 SW 是多少?	What are typical HSQC sweep widths?	§ 4.1
21	间接维混叠有什么坏处和好处?	What are the downside and upside of aliasing in F1?	§ 4.2
22	256 增量跑 12.8 分钟，其它不变时 40 增量要多久?	If 256 increments take 12.8 min, how long do 40 take?	§ 4.3
23	按顺序说出上机的六步。	List the six steps at the spectrometer, in order.	§ 5.1
24	看调谐曲线时，怎么区分调谐和匹配?	On the wobble curve, how do you tell tuning from matching?	§ 5.2
25	Lock 和 shim 各解决什么问题?	What does the lock fix, and what does shimming fix?	§ 5.3, 5.5
26	lock 溶剂设错会怎样?	What happens if the wrong lock solvent is set?	§ 5.4
27	为什么 p1 不准会影响形状脉冲?	Why does a wrong p1 affect shaped pulses?	§ 5.6
28	反转恢复实验的序列是什么? 它测什么?	What is the inversion recovery sequence, and what does it measure?	§ 5.6
29	设 SI = TD 时发生了什么? 这增加了信息吗?	What happens when SI = TD? Does it add information?	§ 6.2
30	补零和线性预测有什么区别? LP 有什么风险?	How do zero filling and LP differ? What is the risk of LP?	§ 6.3
31	后向线性预测主要解决什么?	What does backward LP mainly solve?	§ 6.4
32	PH0 和 PH1 有什么区别?	What is the difference between PH0 and PH1?	§ 6.5
33	基线的三类常见问题，各用什么修?	Name three common baseline problems and their fixes.	§ 6.6
34	LB 的原则是什么? 氢谱和碳谱的常用值?	What is the LB rule, and typical values for proton and carbon?	§ 6.8
35	HSQC 和 COSY 各用什么窗函数?	Which window functions suit HSQC and COSY?	§ 6.9

答案 Answers

1	信号只有微伏级，放得近能把电阻损失降到最低；放大后损耗就不那么要紧。	The signal is only microvolts; close placement minimizes resistive loss. Once amplified, losses matter less.
2	65 536 级；256 级。	65 536; 256.
3	10 kHz（v/2）。更高的频率会混叠成假信号。	10 kHz (v/2). Higher frequencies alias into fake signals.
4	DW = 1/(2×8000) = 62.5 µs；AQ = 32768 × 62.5 µs = 2.048 s。	DW = 62.5 µs; AQ = 2.048 s.
5	波纹；波纹加分辨率下降；只有分辨率下降；定量信息丢失。	Wiggles; wiggles plus lost resolution; lost resolution only; lost quantitation.
6	AQ 变短，分辨率变差。	AQ gets shorter; resolution gets worse.
7	太高失真（ADC 溢出）。太低不失真，但白丢 SNR，小峰受害最大。	Too high distorts (ADC overflow). Too low loses SNR, hurting the smallest peaks most.
8	单扫氢谱下调到畸变消失，或用 rga；涉及碳的实验通常用最大值。	Lower it on a single-scan proton until distortions vanish, or use rga; usually max for carbon.
9	10 s；约 99.3%。	10 s; about 99.3%.
10	不总值得：结构解析常规条件够用；同一分子的定量研究把误差从 8% 降到 1%，却花了 5 倍时间。	Not always worth it: routine conditions suit structure work; quantitation cut error from 8% to 1% but took 5× as long.
11	4 倍（SNR ∝ √NS）。	4× (SNR scales with √NS).
12	SFO1 = BF1 + O1。O1 和 O1P 是同一个值，单位分别是 Hz 和 ppm。	SFO1 = BF1 + O1. O1 and O1P are the same value, in Hz and ppm.
13	看不到峰；带宽不足，激发不均匀；溶剂压不掉。	Missed peaks; uneven excitation from limited bandwidth; solvent not suppressed.
14	90° 的最大值平缓，难判断；360° 是过零点，容易读准。（解释为讲义外补充）	The 90° maximum is flat; the 360° null is sharp and easy to read. (Explanation beyond the slides.)
15	带宽变窄，远处峰强度下降甚至变负。两者成反比，长度只在特定功率下正确。	Narrower bandwidth, far peaks weaken or go negative. Inversely related; a length is only right for a given power.
16	更长、功率低得多，可选择性激发，加热更少。探头扛不住高功率时可能必须用。	Longer, far less power, selective, less heating. Needed if the probe cannot handle high power.
17	zgig：只在 AQ 期间去耦，没有 NOE。	zgig: decouples only during AQ, so no NOE.
18	功率之比，而非绝对值；之后要有恢复延迟，否则谱图畸变。	The ratio of powers, not the absolute value; a recovery delay must follow or spectra distort.
19	1/(4 × 145) = 1.72 ms。	1/(4 × 145) = 1.72 ms.
20	^{1}H 约 20 ppm，^{13}C 约 200 ppm。	About 20 ppm in ^{1}H and 200 ppm in ^{13}C.
21	坏处：和真实峰重叠。好处：折叠到空白区，缩短实验时间。	Downside: overlap with real peaks. Upside: fold into an empty region to save time.
22	12.8 × 40/256 = 2.0 分钟。	12.8 × 40/256 = 2.0 min.
23	样品 → 调谐匹配 → 锁场 → 匀场 → 校准脉冲 → 设参数。	Sample → tune and match → lock → shim → calibrate → set parameters.
24	调谐看凹陷的位置是否对准中线；匹配看凹陷够不够深。	Tuning: is the dip on the centre line? Matching: is the dip deep enough?
25	Lock：磁场随时间的稳定性。Shim：磁场在样品空间内的均匀性。	Lock: stability over time. Shim: homogeneity across the sample.
26	可能锁不上，或化学位移全部错误。	It may not lock, or chemical shifts may be wrong.
27	形状脉冲不单独校准，而是根据 p1 算出来的。	Shaped pulses are not calibrated separately; they are calculated from p1.
28	180° — t — 90°，改变 t，拟合得到 T1。	180° — t — 90° with varying t; the fit gives T1.
29	实际点数翻倍（补零），数字分辨率提高，但没有新信息。	The points double (zero filling), digital resolution improves, but no new information.
30	补零只加 0；LP 外推 FID。LP 加的是“假”数据，可能带来伪影。	ZF just adds zeros; LP extrapolates the FID. LP adds “fake” data and can create artifacts.
31	声学振铃造成的基线差；也有助于调整群延迟。	Poor baselines from acoustic ringing; it also helps with group delay.
32	PH0 给每个峰加同样的量；PH1 的量取决于峰到支点的距离。	PH0 adds the same to every peak; PH1 depends on the distance from the pivot.
33	Smile/frown：线性校正。整体弯曲：多项式。宽探头背景：可校正但难得多。	Smile or frown: linear. Overall curve: polynomial. Broad probe background: possible but much harder.
34	小于最尖峰的线宽。^{1}H 0–0.5 Hz；^{13}C 5–15 Hz。	Below the sharpest peak’s width. ^{1}H 0–0.5 Hz; ^{13}C 5–15 Hz.
35	HSQC：两维都用移位 sine²。COSY：不移位 sine²。	HSQC: shifted sine² in both dimensions. COSY: unshifted sine².